



PROJECT DOCUMENTATION / PART 2 OF 2

System, interfaces and operations

The complete application reference: files, APIs, telematics, Google Sheets, data contracts, every page, maps, messages, storage, deployment, testing and extension points.

Prepared for Div
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Archive: Fuelio-main.zip

This manual describes the supplied implementation. It separates implemented behavior, verification evidence, known limitations, and suggested future work. No source files or deployed services were changed.

Archive SHA-256

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Contents

1. Project identity and architecture7

1.1 Technology stack7

1.2 Where work happens8

1.3 Shared-link access8

2. Source modules and application startup9

2.1 Module responsibilities9

2.2 Startup sequence9

2.3 Navigation and shared state9

2.4 Protected state transitions10

3. The Master workbook and Google connection11

3.1 One registry workbook, several linked catalogs11

3.2 Fuel and message registry row rules11

3.3 Client rows and multiple Samsara tokens11

3.4 Current mapping behavior12

3.5 Service-account authentication12

3.6 Public fallback and partial configuration12

4. Fuel-stop data, aliases and import behavior13

4.1 Standard fuel template13

4.2 Complete accepted fuel header aliases13

4.3 Normalised stop record14

4.4 Invalid rows and duplicates15

4.5 Numeric, state and highway normalization15

4.6 CSV and XLSX details15

5. Message catalogs and Sheet instructions16

5.1 Standard message template16

5.2 Accepted message layouts16

5.3 Direct Sheet references16

6. Samsara: endpoints, aggregation and telemetry	17
6.1 Current provider support	17
6.2 Complete external Samsara call inventory	17
6.3 Bounded calls, retries and pagination	17
6.4 Combining connections and choosing the latest stats	18
6.5 Public vehicle data contract	18
6.6 Vehicle MPG	19
6.7 Browser detail hydration	19
6.8 Editable planning values and Use live	19
7. Internal API reference	20
7.1 Common request and error behavior	20
7.2 All public application endpoints	20
7.3 Configuration and source request schemas	21
7.4 Routing request schemas	21
7.5 Generic outbound fetch helper	21
7.6 Exact external destinations in this snapshot	21
8. Planner page and result workflow	24
8.1 Layout and pinned actions	24
8.2 Inputs and vehicle selection	24
8.3 Planner map before and after route finding	24
8.4 Find routes and Fuelio Hunt	25
8.5 Result cards and actions	25
8.6 Reset behavior	25
9. Fuel Stops page: complete interaction contract	26
9.1 Shared records and list/map layout	26
9.2 Live search and ranking	26
9.3 State filter	26
9.4 Multi-select highways	26
9.5 The map/filter rule	27
9.6 Price cycle and export	27

9.7 List-to-map and map-to-list selection	27
9.8 Stop details	27
9.9 Clear and Fit	27
10. Fleet page: complete interaction contract	28
10.1 Fleet selection and visibility	28
10.2 List data and detail hydration	28
10.3 Vehicle icon and information	28
10.4 Fleet-only control differences	28
11. Messages: one editor in two places	29
11.1 Standalone and Planner parity	29
11.2 Manual identifiers and resolution	29
11.3 Stop pairs	29
11.4 Planner autofill	29
11.5 Primary-driver template	30
11.6 Team-driver template	30
11.7 Preview editing, persistence and limits	31
12. Maps, Hybrid references and Price Heat	32
12.1 Leaflet base and layer order	32
12.2 Primary Hybrid vector style	32
12.3 Load, readiness and fallbacks	32
12.4 Price Heat is a price colour scale	33
12.5 Fit, focus and empty cases	33
13. Themes, branding, responsiveness and accessibility	34
13.1 Four page themes	34
13.2 Branding and vehicle assets	34
13.3 Layout breakpoints	34
13.4 Accessibility features and limits	35
14. Persistence, refresh and cache reference	36
14.1 Browser preferences	36
14.2 IndexedDB data	36

14.3 Backup and restore 36

14.4 Complete timing/cache table 37

14.5 What live every minute really means 37

15. Activity, saved routes and hidden load tools 39

15.1 Activity 39

15.2 Retained hidden workflows 39

15.3 Complete bundled route anchors 39

15.4 Load template and alias map 40

16. Build and deployment operations 41

16.1 Repository and scripts 41

16.2 Public build contents 41

16.3 Vercel configuration 41

16.4 All environment variables read by the application 42

16.5 Credential replacement procedure 42

16.6 Hosting and external-service limits 43

16.7 Updating and rolling back 43

17. Diagnostics, privacy boundaries and failure recovery 44

17.1 Diagnostic ladder 44

17.2 Connection truthfulness 44

17.3 Credential/data separation 44

17.4 Source and data quality responsibility 45

18. Testing, integrity and evidence 46

18.1 Documentation verification result 46

18.2 Calculation and parsing tests 46

18.3 Page and map tests 46

18.4 Backend integration and fault tests 46

18.5 Not verified by this run 46

19. Current gaps and reconciliation with earlier plans 48

19.1 Implemented versus discussed 48

19.2 Documentation drift inside the repository 48

19.3 Maintenance risks visible in the source 49

20. Extension guide, including other telematics providers 50

20.1 What can remain stable 50

20.2 Proposed adapter design - not implemented 50

20.3 Editable client-price mapping - not implemented 50

20.4 Change-impact map 51

20.5 Optional browser model-context tools 51

21. Reference and handoff notes 52

21.1 Reading the two volumes together 52

21.2 External primary references 52

21.3 Handoff acceptance checklist 52

A. Complete file inventory and integrity 54

B. Complete dependency lock inventory 57

C. Runtime function and class locator 66

Class methods and compact shared helpers 71

D. Full HTML control inventory 72

E. Complete test case index 75

1. Project identity and architecture

Fuelio is a browser-based dispatch application for an internal team working with multiple client fleets. Its visible work areas are Plan trip, Fleet, Fuel Stops, Messages and Activity. A retained Routes & loads area is hidden. The app uses a Google Master workbook to discover fuel catalogs, message catalogs and Samsara client connections, then reads the chosen data through a server proxy. It also supports manually entered trips and uploaded data when telemetry is unavailable.

This manual documents the complete supplied Fuelio-main.zip snapshot, including infrastructure, data contracts, source modules, interfaces, maps, messages, storage, tests and extension points. The separate Part I explains the optimiser, route finder and every fuel calculation. This is documentation of the current implementation, not an implementation of new requested features.

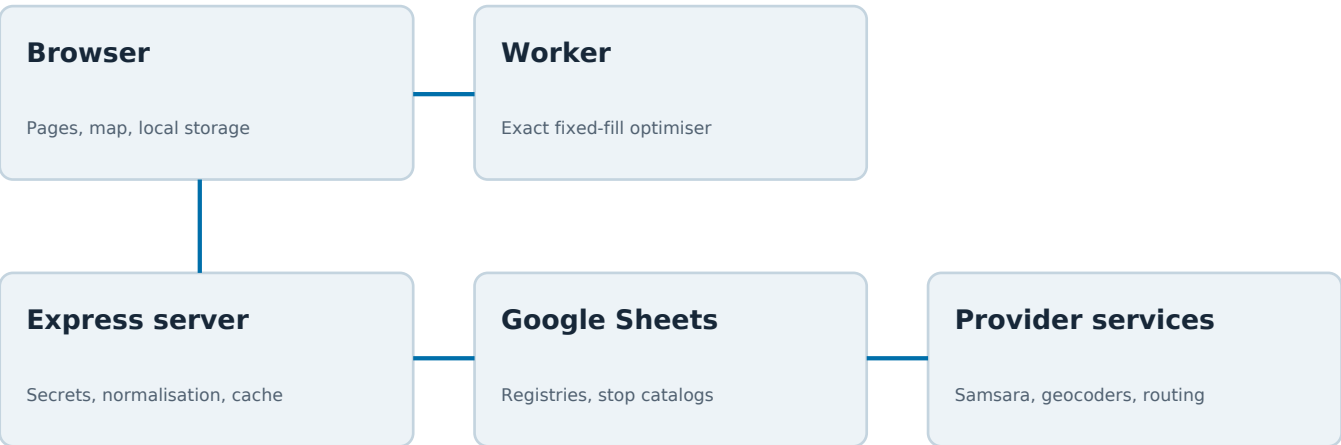
1.1 Technology stack

Component	Implementation
Application UI	Plain HTML, CSS and JavaScript ES modules; workspace.js coordinates the views.
Primary map	Leaflet 1.9.4; one shared map with independent layers and view-specific data.
Hybrid references	MapLibre GL 4.7.1 through @maplibre/maplibre-gl-leaflet 0.0.22.
Spreadsheet files	ExcelJS 4.4.0 for .xlsx in the browser; custom CSV parser for delimited files.
Server	Express 5.1.0, Node ES modules, default application export for Vercel.
Optimisation	Custom exact full/half-fill dynamic programme in a browser Web Worker.
Persistence	Browser localStorage and IndexedDB; server memory caches.
Tests	Node test runner and JSDOM 26.1.0, with provider/map fixtures.
Deployment	GitHub-connected Vercel Express project; static assets generated into public/.

The package's internal name is fco-dispatch and version is 1.0.0. The visible product name is Fuelio. Some storage keys, download filenames, internal variable names and health/version labels retain FCO or pump naming for compatibility. The UI wording has been changed to Stop/Stops where the updated interface uses it; a legacy pump_name import header remains intentionally valid.

There is no React, Next.js, Streamlit, Python service, database server, Google Maps SDK, OR-Tools or commercial solver in this archive. PurFCO is the historical origin of route-library and fuel-balance ideas, not a separately running Python application inside Fuelio. The old Samsara Python SDK ZIP is not a runtime dependency of this project.

1.2 Where work happens



Browser also loads imagery, vector references and fonts directly from their providers.
Google and Samsara credentials remain on the server; the optimiser has no network access.

The browser receives normalised vehicle and source records, never the configured Google private key or Samsara bearer tokens. The worker receives only numeric planning inputs and candidate stop data. It does not fetch telemetry, maps or routes. The server validates source references, manages credentials, performs external calls, and returns the data shapes that the UI consumes.

Map tiles, vector styles/tiles, glyphs, sprites and Google Fonts are separate browser requests to their publishers. A functioning /api/health response does not prove that those visual providers, Google Sheets, Samsara or route providers are available.

1.3 Shared-link access

There is no application login, user role system or per-client access authorisation in the source. The intended shared-link experience lets an operator select any configured client. Client IDs identify which fleet to read; they are not passwords. The server rejects requests explicitly marked cross-site by the browser, but that check is not authentication and does not make a public URL private.

This distinction matters when maintaining the app: hiding a tab is a presentation choice, and withholding API keys from the browser is a credential boundary. Neither creates user permissions. The document preserves the requested shared-link model without claiming tenant-level security that is absent from the code.

Source: package.json; index.html; build.js; server.js; workspace.js; worker.js.

2. Source modules and application startup

2.1 Module responsibilities

workspace.js is the central controller. It reads and persists form state, loads registries and fleets, coordinates data imports, constructs the map, binds UI events, performs route/plan requests, renders results, handles messages and owns hidden library/batch workflows. Because it contains many responsibilities, a behavior change often needs both a focused helper and its workspace integration.

data.js supplies common text, numeric, CSV, header, state, highway and source parsers. view-model.js contains pure stop-search/map-filter logic, active-vehicle filtering, heading markup, report merging and route-highway extraction. place-search.js owns the reusable address autocomplete lifecycle.

message-editor.js owns a reusable, independently editable composer; messages.js supplies its text format and a retained older quantitative message helper. hybrid-map.js owns reference-map style transformation and fallback lifecycle.

optimizer.js, geo.js and batch.js contain the calculation and route-preparation helpers detailed in Part I. worker.js is a small bridge to solveFuelPlan. server.js is the only server-side API application. build.js copies an explicit list of public files and vendor assets. index.html supplies static elements; styles.css supplies themes and responsive layout, including later override blocks.

The appendices inventory every file, every locked dependency path, the top-level runtime functions and the HTML controls. Their line numbers refer to this exact archive.

2.2 Startup sequence

The module first reads fco.workspace.v1 from localStorage, tolerating missing or malformed JSON by starting with defaults. It constructs an application state object with empty vehicles, routes, selected route IDs, plans, history, loads and logs, plus restored client/master choices, overrides, map mode, Price Heat and stop-view settings. It opens IndexedDB fco-workspace version 1 and creates a single data object store on first use.

bootstrap restores allowed form values, applies the theme, restores stop-search/sort controls, initialises both message editors, binds events, reconstructs waypoint rows and creates the map. It reads fuel, saved routes, history, loads, road-access cache and message draft from IndexedDB while fetching route-library.json. A failed library fetch becomes an empty array rather than blocking the visible planner.

Stored fuel and messages are applied before live requests. The URL hash chooses the visible page if it is recognised; otherwise Planner opens. The app then loads /api/config and the selected fleet. A failure sets a manual/unavailable connection status and logs the problem while leaving the workspace usable for manual input.

After startup it installs a 60-second refresh interval and a visibility-change refresh when the tab becomes visible. A running refresh flag prevents overlapping entire scheduled refresh cycles. The timer is in the browser, not a Vercel scheduled function.

2.3 Navigation and shared state

Recognised hashes are planner, fleet, pumps, stops, messages and activity. The older pumps hash and newer stops label both lead to the stop view. Routes & loads is omitted from allowed navigation and redirects to Planner if requested. The brand links to Planner. Switching pages preserves the active source and shared application state; it does not create an independent fleet or map instance.

The map's stop, route, vehicle and connector layers are rendered differently by page. The Fleet page uses truck data and hides stop-specific controls. Fuel Stops uses the viewer filter rules. Planner combines fuel stops with active vehicles, then isolates the planning truck after a route/plan is present. Messages and Activity focus on their own content and do not expose an unrelated pump-count overlay.

2.4 Protected state transitions

Fleet and fuel requests use sequence counters and source/client context checks so a late response from an old selection cannot overwrite a newly selected client/master. Vehicle detail requests have separate deduplication. Route requests also compare their starting trip stamp on completion. An active solve owns an AbortController and worker lifecycle.

These controls prevent common asynchronous UI races, but they do not make all operations transactional. An older successful plan can remain visible during a failed refresh, and a source selector change can occur before its data successfully loads. The visible loaded-source badge and each plan's stored source are therefore meaningful evidence of which data are in use.

Source: workspace.js:1-240, 369-448, 471-533, 685-762, 2278-2498.

3. The Master workbook and Google connection

3.1 One registry workbook, several linked catalogs

The default Master spreadsheet ID is 1VpTN7OY40d1yVFLvtBu2otlbUjAjqm6DVy71FDsBeeY. Its three registry tabs identify other sheets and client connections. The fuel registry is not itself the full fuel-price table unless its rows point to such data. This two-level design lets the user add masters or clients without rebuilding the frontend.

Registry	Default GID	Content
Fuel Stops registry	0	Master Name, Google Sheet URL, GID, Active.
Driver Message Masters registry	970811620	Master Name, Google Sheet URL, GID, Active.
Samsara client registry	1100093231	Name plus Samsara API, Samsara API 2, Samsara API 3 and further numbered token columns.

Environment variables can replace the Master ID and these three GIDs. A GID is a numeric tab identifier; it is not a worksheet index or a tab name. The server resolves the current tab title from spreadsheet metadata. Renaming a tab therefore does not break a valid GID reference. Each fuel/message registry row supplies the GID of its own linked workbook, independent of the three registry GIDs above.

3.2 Fuel and message registry row rules

Headers are normalised by lower-casing and removing non-alphanumeric characters. Master names accept mastername or name. Sheet links accept googlesheeturl, sheeturl or url. GID is parsed separately. A row without a name or URL is skipped. An invalid nonblank Sheet reference can fail that registry load, rather than silently loading a different workbook.

Active is considered enabled when missing, blank, null, or one of true, yes, 1 and active after lower-casing. The active-value function does not trim arbitrary surrounding spaces, so TRUE is reliable whereas a cell containing spaces around TRUE can behave differently. Use an actual boolean or clean TRUE/FALSE cell.

Master IDs are 24 hexadecimal characters derived from SHA-256 of name, spreadsheet ID and GID. Renaming a master or changing its source changes its ID and can invalidate a stored selection. An active registry list is live data; the application does not hardcode a maximum number of clients or masters.

3.3 Client rows and multiple Samsara tokens

A client row uses Name or Client. Normalised headers matching samsaraapi followed by zero or more digits are read as token columns. Blank token cells are ignored and repeated identical tokens in a row are deduplicated. Thus one client can have one connection while another has two or more. The server fetches the connections and combines their vehicles for that selected client.

The client ID is a hash of its name. Names should be unique: duplicate names can create the same ID and an ambiguous lookup. Changing a name changes the ID and therefore its local selection/override identity. The public configuration sends only client ID, name and connection count, never token cell values.

If no clients are found, an optional server environment fallback reads SAMSARA_TOKEN, separated by commas or actual newline characters. SAMSARA_CLIENT_NAME supplies its display label, defaulting to Connected fleet. This fallback is optional; the normal setup uses the private Master workbook.

3.4 Current mapping behavior

The current source does not enforce a client-to-fuel-master mapping. Client and fuel master are independent, editable selections. Changing a client does not automatically change the selected price master, and /api/fuel does not check client membership.

On a fresh browser, the first available client/master can be selected. There is no permanent Young's default in the current code. Browser persistence remembers choices later. The displayed master name/count and the plan's stored source help the user recognise the selected data, but they do not prevent cross-client price selection. An explicit editable mapping was discussed earlier; it remains a future change rather than an implemented control in this archive.

3.5 Service-account authentication

GOOGLE_SERVICE_ACCOUNT_JSON contains the complete JSON credential server-side. The parser requires client_email and private_key. The server creates an RS256-signed JWT with the service-account email as issuer, the Google Sheets read-only scope, Google's token endpoint as audience, current issue time and a one-hour expiration. Escaped private-key newline sequences are converted for signing.

The JWT is exchanged using a form-encoded OAuth token request with a 15-second timeout. The resulting access token is cached for 50 minutes. Spreadsheet metadata is cached for ten minutes. To read a tab, the server maps GID to title, safely quotes the A1 sheet name and calls the Sheets values endpoint with UNFORMATTED_VALUE. It reads rows; it does not write prices, tokens, routes or messages back into Sheets.

Share the registry workbook and every private linked fuel/message workbook with the service account's client_email as Viewer. Replacing a key on the same service account normally keeps that email; creating a different service account requires sharing with its new email. A pasted key being syntactically valid does not establish that the account has access to the required workbook.

3.6 Public fallback and partial configuration

Without a configured Google credential, public fuel/message sheets can be read through Google's gviz CSV export. The Samsara token registry is privateOnly and never uses a public export. A configured private credential that fails is not silently bypassed with a public read. A workbook holding API tokens should remain Restricted.

registry reads its three tabs using Promise.allSettled. It can return available masters while reporting a client-tab failure, or preserve clients while a message registry is temporarily unavailable. The response includes errors and per-section available flags. Browser refresh retains the previous section when the corresponding availability flag is false. privateMaster indicates that private credentials are configured, not an independent check of every sheet's sharing policy.

The registry cache lasts 55 seconds. Changes are picked up through normal refresh after cache expiry; there is no instant push from Sheets, automatic workbook editing or central transactional configuration database.

Source: server.js:112-370; workspace.js:369-448; ENVIRONMENT.txt. Google OAuth/Sheets credentials were not printed or tested against production during this review.

4. Fuel-stop data, aliases and import behavior

4.1 Standard fuel template

The supplied fuel-template.csv contains only its header row. It defines the eight recommended columns below. The updated Fuel Stops interface uses Stop terminology, while pump_name remains the stable import header. Keep numeric prices and coordinates as numbers; do not add unit text to numeric cells.

Column	Meaning / example
pump_name	Stop identity, such as Pilot - IN30 or Love's #245.
price_per_gallon	Positive US-dollar price per US gallon, such as 4.1234.
latitude	Decimal latitude, such as 39.82150284.
longitude	Decimal longitude, such as -85.91623097.
city	City name.
state	State abbreviation or a recognised full state name.
highway	Original road text, such as I-70.
Exit	Exit label, such as 96 or 107/24th St.

The parser actually requires recognised price, latitude and longitude headers. A name/store header adds confidence to header detection but is not mandatory. If no usable name exists, it derives one from brand/store or falls back to a row-based name. This is broader than the older README statement that the first four columns are mandatory.

4.2 Complete accepted fuel header aliases

Spaces, punctuation, underscores and letter case are removed before matching. These lists are ordered: when several columns map to a field, the first alias found in the list wins. In particular, the parser chooses a preferred discounted/net price column by alias priority, not the minimum number across all price columns in a row.

Normalised field	All accepted normalised aliases, in priority order
name	pumpname, truckstop, stationname, pump, name, storename, locationname, station, fuelstop, sitename, vendor, store, columna, a
store	storenumber, storeno, store, pumpnumber, stationid, storeid, id
price	bestdiscountedprice, discountedprice, netprice, pricepergallon, pricepergal, pricegallon, costpergallon, cpg, dieselprice, price, fuelprice, bestprice, retailprice, rate, columnb, b
lat	latitude, lat, latitudedegrees, y, columnc, c
lng / lon column	longitude, long, lng, lon, longitudedegrees, x, columnd, d
city	city, town, municipality, columne, e
state	state, statecode, province, st, columnf, f

Normalised field	All accepted normalised aliases, in priority order
location	location, address, citystate, fulladdress
brand	brand, network, chain
highway	highway, hw, interstate, route, highways, highwayname, interstatehighway, corridor, routename, road, columng, g
exit	exit, exitnumber, exitno, exitname, columnh, h
mile	distancefromstartmiles, routemile, mile, milepost

Header detection scores the first 12 rows: three points each for a recognised latitude, longitude and price, plus one for name/store. The highest scoring row is selected; the minimum acceptable score is nine. Introductory title rows are therefore supported only within that scan window. Equal best scores keep the earlier row.

A legacy generic five-column A-E or Column A-E layout can be interpreted as name, location, price, latitude, longitude when the sample indicates a text second column and numeric third column. This compatibility branch is explicit and should not be mistaken for arbitrary column inference.

4.3 Normalised stop record

Field	Source / transformation
id	Lower-cased brand, store or lower-cased name, and coordinates rounded to six decimals.
name	Original name; numeric names may become brand plus #store.
store	Explicit store field or matching trailing numeric identifier.
brand	Explicit value; otherwise inferred for Love's, Pilot or Flying J from name/source text.
price	Positive parsed numeric price.
lat / lng	Numeric decimal coordinates.
city / state	Trimmed city and normalised state.
location	Original location or city/state combination.
highway	Original trimmed highway text.
highways	Deduplicated recognised I-/US-/SR- tokens.
exit	Original trimmed text; blank stays blank.
source	Display name of the selected source.
mile	Optional parsed imported mile value; route projection later supplies planning progress.

Both the stop table and map use this same record set. No second price-normalisation path exists for map markers. A stop's geographic identity includes its coordinates, so two differently located stations with the same store number are not automatically collapsed. Conversely, very close coordinates rounded to the same six-decimal identity can collapse.

4.4 Invalid rows and duplicates

Blank body rows are ignored. Missing/non-numeric coordinates, latitude outside -90 to 90, longitude outside -180 to 180, missing price or price at/below zero produce a row issue and are skipped. Geographic validity alone accepts coordinates such as 0,0; subsequent US routing checks can still reject them. If no valid rows remain, loading fails.

Identical normalised IDs with prices within 0.00000001 are omitted as duplicate rows and reported in issues. Conflicting prices for the same ID abort the entire parse so the app cannot arbitrarily pick an entitlement. Other differing descriptive fields on a same-price duplicate do not create a second row. Result metadata includes rows, issues, source and loadedAt. loadedAt is when the app parsed the data, not when the supplier last changed the price.

4.5 Numeric, state and highway normalization

number accepts finite numeric values and trimmed decimal strings with optional sign. It removes dollar signs, percent signs and whitespace, and recognises conventional comma thousands groups. It rejects exponent strings, arbitrary unit suffixes and empty values. A percent symbol is removed without dividing by 100; the calling form handles percent-to-gallons conversion.

stateCode maps exact US state/DC names or abbreviations case-insensitively and otherwise upper-cases the original text. This import behavior remains active even though the search feature no longer expands California, Cal or Cali into CA. These are separate behaviors.

highwayTokens normalises Interstate to I-, and recognises I, US and SR followed by digits and an optional trailing letter. It can extract multiple such tokens from a cell. State-specific strings such as CA-46 remain in raw text but are not necessarily selectable tokens. A combined I-85/77 string yields the recognised I-85 portion; it does not infer a second Interstate automatically. Missing highway/exit text is never invented from coordinates.

4.6 CSV and XLSX details

parseCSV removes a UTF-8 BOM and guesses tab, comma or semicolon delimiters by frequency across the first 12 text lines. The guess is a heuristic; it does not first strip quoted content. The character parser handles quoted delimiters, doubled quotes, embedded line breaks, CRLF and empty rows. An unclosed quoted field is an error.

CSV export quotes every cell, doubles embedded quotes and uses CRLF rows. It prefixes an apostrophe for certain spreadsheet-formula-leading strings, while preserving numeric negative values. This is a limited export safeguard, not a full spreadsheet-security parser for every leading-whitespace/formula case.

readTableFile permits files up to 15,000,000 bytes. .xlsx loads the bundled ExcelJS library on demand. If multiple eligible sheets exist, the user selects one; veryHidden sheets are excluded, while ordinary hidden sheets can remain eligible. A worksheet over 100,000 rows is rejected after loading. Date values become ISO text; rich-text fragments are concatenated; formula cells use cached result values. Fuelio does not evaluate spreadsheet formulas or support the older binary .xls format. Large compressed workbooks can consume memory before the row-count guard.

Source: data.js:1-347; workspace.js:748-762, 1883-1952; fuel-template.csv.

5. Message catalogs and Sheet instructions

5.1 Standard message template

The Messages page and Planner message dialog share a Download CSV template action and Google Sheet template instructions. The supplied message-template.csv has these headers and three illustrative rows. Coordinates are retained in the template for compatibility with the original viewer, but message composition uses the textual stop identity and location/highway/exit.

StoreNumber	Latitude	Longitude	Location	Highway	Exit
245	35.377833	-97.573646	Oklahoma City, OK	I-44	113
211	35.377871	-97.475970	Oklahoma City, OK	I-35	120
88	(blank)	(blank)	Yukon, OK	(blank)	(blank)

To prepare a catalog, place those headers in the first row, enter one station per row, keep StoreNumber identifiers consistent and leave unavailable Highway/Exit values empty. Upload CSV/XLSX, use a selected registry master, or provide a supported Google Sheet URL and the intended GID. Private linked sheets must be shared with the configured service-account email. Public read access is only an alternative for non-secret fuel/message data when appropriate; it is never the way to expose a token registry.

5.2 Accepted message layouts

parseMessageTable looks through the first 12 rows for a store-number header plus a recognised location/city/address/highway header. It uses the shared fuel-field alias lists for store, brand, location, city, state, highway and exit. It also recognises headerless four-column rows as store, location, highway, exit, and headerless six-column rows as store, latitude, longitude, location, highway, exit.

Unlike fuel-stop parsing, message parsing does not require a price or valid coordinates. It produces id, store, brand, location, highway, exit, name and source. A brand comes from an explicit column or the source naming logic. The parser does not deduplicate a whole catalog; the editor resolver deals with ambiguity when the user types an identifier.

Use distinct master labels such as Love's and Pilot to make the active message catalog obvious. The selected message master has its own indicator and can differ from the currently selected price master. A message catalog enriches driver directions; it does not determine the price used by the optimiser.

5.3 Direct Sheet references

parseSheetReference accepts a spreadsheet ID or a docs.google.com/spreadsheets/d/... URL. It rejects arbitrary non-Google hosts. The GID comes from the query, hash or zero default, with an explicit override taking priority even when its value is zero. The GID must be numeric. The row's displayed master name becomes source metadata.

This parsing does not grant read permissions. A correct URL with a missing share still fails at Google. The Sheets service account can access only documents shared with it or otherwise accessible to it; Fuelio does not search the operator's whole Drive account or choose a worksheet by guesswork.

Source: data.js:317-398; message-editor.js; message-template.csv; server.js:341-370.

6. Samsara: endpoints, aggregation and telemetry

6.1 Current provider support

Samsara is the only implemented telematics provider. The base host, bearer authentication, inventory endpoints, stats fields and fuel-report response interpretation are in `server.js`. Putting a different provider's API key into a Samsara API cell will not connect it. Google Sheets supplies credentials and client grouping; it does not translate API formats.

The server intentionally requests snapshots, not a continuous stats feed. It separates basic fleet location/fuel from optional detailed MPG/history work so a slow report does not need to block every truck's location. The frontend separately hydrates report details and preserves missing values rather than displaying unknown fuel as zero.

6.2 Complete external Samsara call inventory

Endpoint	Request / purpose	Scope and retention
GET /fleet/vehicles	Inventory, limit=512, cursor pagination	Per token; cached for ten minutes.
GET /fleet/vehicles/stats	types=gps,fuelPercents,obdOdometerMeters	Current snapshot group; paginated.
GET /fleet/vehicles/stats	types=gpsOdometerMeters,engineStates	Separate snapshot group; paginated in parallel with the first.
GET /fleet/reports/vehicles/fuel-energy	vehicleIds plus a previous-30-day startDate/endDate	Optional vehicle MPG calculation.
GET /fleet/vehicles/stats/history	vehicleIds, types=fuelPercents and previous-seven-day startTime/endTime	Latest known fuel fallback when current fuel is missing.

The exact endpoint response fields and permission behavior must be supplied by Samsara. The local integration tests exercise these contracts using fixtures, not the user's production account. The app does not call the historical Python SDK, driver messaging APIs, engine diagnostics APIs, trip-dispatch write endpoints or webhook subscriptions.

Reference: [Samsara stats snapshot](#) and [fuel/energy vehicle reports](#). The table describes the calls present in `server.js:399-689`.

6.3 Bounded calls, retries and pagination

`samsaraRead` sends an Authorization bearer header, rejects redirects and applies a 12-second attempt deadline that covers the response body as well as headers. It permits two attempts for transient network errors, 429 and server failures, with a 500 ms retry wait. This Samsara-specific retry path does not use the generic fetch helper's Retry-After handling. Authentication rejection is explained without returning the token.

`samsaraPage` appends cursor values while `pagination.hasNextPage` is true and collects either the normal data array or supported `vehicleReports` data shape. It errors if a next cursor is missing or repeats and caps the loop at 500 pages. Those guards prevent malformed provider pagination from running forever.

`deadline` races the whole task against a timer, propagates a parent abort signal and cleans up timers/listeners. The fleet setup includes a 60-second client-registry lookup deadline. Each per-token fleet operation has a 45-second total budget, and the optional stats groups each have a 25-second budget within

it. These budgets include pagination and retries, unlike a timeout attached only to the initial response headers.

6.4 Combining connections and choosing the latest stats

For a selected client, its tokens are processed with `Promise.allSettled`. One valid connection can still produce a partial fleet when another fails. Within a token, an inventory failure fails the connection; a stats-group failure produces warnings and can retain previously known stats for inventory vehicles. The merged per-token result is cached for 50 seconds.

Vehicles are combined by Samsara vehicle ID. Metadata from the earlier encountered vehicle record is retained, while each stat field is merged using the newest measurement timestamp. GPS, fuel and odometer can therefore come from different samples/times, which is preferable to replacing an entire vehicle with an older all-field object. Arrays of stat observations are reduced to their latest sample.

The server keeps per-token last-stat state in memory, pruning vehicles that disappear from that token's inventory and capping the number of stored token maps at 100. This is not durable history. A cold start or new server instance does not share that retained cache. Vehicles with the same name or VIN but different IDs are not automatically deduplicated. The same ID can exist in different selected clients without their whole fleets being combined across client boundaries.

6.5 Public vehicle data contract

Field returned to browser	Derivation and null behavior
id, name	Inventory identity and display name.
vin, make, model, year, licensePlate	Inventory metadata, when provided.
lat, lng	Latest GPS latitude/longitude parsed numerically; missing stays null.
location	GPS <code>reverseGeo.formattedLocation</code> , else <code>GPS address.name</code> , else blank.
gpsTime	GPS sample time, else null.
speed	GPS <code>speedMilesPerHour</code> , else null.
heading	GPS <code>headingDegrees</code> , else null.
fuelPercent	Latest <code>fuelPercents</code> value, else null.
fuelTime	Fuel sample timestamp, else null.
odometer	OBD odometer preferred, otherwise GPS odometer; metres divided by 1,609.344.
odometerTime	Timestamp of the chosen odometer sample, else null.
engine	Latest <code>engineStates</code> value, else null.

The fleet response additionally reports warnings, whether it is partial, `connectionsLoaded`, `connectionsTotal` and `fetchAt`. Sample timestamps describe measurement age; `fetchAt` describes the application fetch. They are not interchangeable. A successful HTTP response with missing fuel is not evidence that the truck has no fuel.

6.6 Vehicle MPG

GET /api/vehicle-detail accepts client and vehicle identifiers. It verifies inventory membership through the client's token fleet before requesting the report. Vehicle IDs must match a restricted alphanumeric/underscore/colon/hyphen form and be 1-120 characters. The details result is cached server-side for 55 seconds by client and vehicle.

The previous 30 days of distanceMeters and fuelConsumedMI are preferred for MPG. Distance is converted to miles and consumed fuel to US gallons, then divided. For example, 1,000 miles and 125 gallons yield 8 MPG. If that calculation is unavailable, a valid efficiencyMpge field can be used. Values at/below zero or over 100 are rejected. mpgSource labels the report period/source; the app does not pretend that a 30-day average is instantaneous efficiency.

If the current token's fuel percentage is missing, a seven-day fuel-history request can supply the latest known sample. historyFuel is a fallback measurement, not fuel bought, a detected refuelling event, a tank sensor calibration or consumption-adjusted current fuel. It may be old. Current snapshot fuel remains authoritative when available. The result includes mpg, mpgSource, historyFuel and warnings; failure leaves fields null and manual input usable.

6.7 Browser detail hydration

The frontend has a detail cache and a pending-request map keyed by client plus vehicle ID. Basic fleet data can appear first, with MPG and historical fuel arriving later. Background hydration processes two vehicles concurrently, prioritising selected/focused vehicles. The current implementation queues active fleet records rather than implementing true viewport-only list hydration.

When fuel or MPG is missing, the frontend retries details on a shorter roughly 55-second freshness window; complete cached details can be reused for 15 minutes. Selected-vehicle refresh can force a newer detail check. Results are merged into the display without changing the original server schema. A missing report value does not overwrite a valid snapshot fuel with zero.

6.8 Editable planning values and Use live

The planning form can be used without Samsara. Origin, capacity, MPG, start fuel and fuel percent can be manually edited. Overrides are keyed to client/vehicle identity or the manual truck text. Changing start gallons removes a conflicting percent override, and vice versa. Capacity determines conversion between gallons and percent; it is not inferred from Samsara's inventory response in this code.

Use live clears the stored override for that vehicle and applies the currently available live values; the user can edit again immediately. Focused input protections avoid replacing text while it is being edited. An unavailable reading remains distinguishable from a default form assumption: a new selected vehicle may use the form's MPG default if no report value is supplied, while the telemetry record itself can still show missing MPG.

The compact vehicle pane displays truck, speed, fuel percent and MPG. A manually edited compact speed is a display override, not a write to Samsara and not a dynamic fuel-consumption input. VIN, odometer and engine metadata remain available in the data model even though the compact Planner card no longer displays every original field.

Source: server.js:372-689; view-model.js; workspace.js:449-684. Part I describes how the resulting numeric MPG/fuel feed the optimiser.

7. Internal API reference

7.1 Common request and error behavior

The frontend calls same-origin `/api` paths using JSON for POST bodies. The browser helper uses a supplied cancellation signal when one exists; otherwise it creates a 180-second timeout signal. It does not combine both, so a request with its own controller does not also inherit that default browser timeout. It reads the response as JSON and reports an error for a non-success HTTP status. The action wrapper disables the triggering control while running, reports failures and restores the control afterward.

The server uses Express JSON parsing with a 2 MB body limit. API responses set `Cache-Control: private, no-store` and `X-Content-Type-Options: nosniff`. A `Sec-Fetch-Site` header equal to `cross-site` yields a 403 JSON error. This protects some browser cross-site requests, not unauthenticated direct callers. There is no generic open CORS policy or session login in this app.

Unknown API routes return 404. Most thrown application errors become status 400 with an error message; oversized JSON bodies become 413. A fleet with every connection failed returns 503. External provider failures do not uniformly preserve their upstream HTTP code in the public response. A Vercel function timeout can instead produce a platform 504 outside the normal JSON handler.

7.2 All public application endpoints

Method / path	Input	Successful output
GET <code>/api/health</code>	None	<code>ok=true</code> and <code>version=1.0.0</code> . No upstream probe.
GET <code>/api/config</code>	None	<code>masters</code> , <code>messageMasters</code> , <code>clients</code> without tokens, errors, available flags, <code>privateMaster</code> , <code>truckRouting</code> , <code>refreshSeconds=60</code> .
POST <code>/api/fuel</code>	<code>masterId</code> , or url with optional <code>gid</code> and <code>label</code>	Normalised rows, issues, source, <code>loadedAt</code> .
POST <code>/api/messages</code>	<code>masterId</code> , or url with optional <code>gid</code> and <code>label</code>	Normalised message rows and source.
GET <code>/api/fleet</code>	client query	Normalised vehicles, warnings, partial flag, connection counts and <code>fetchAt</code> .
GET <code>/api/vehicle-detail</code>	client and vehicle queries	vehicle ID, mpg, <code>mpgSource</code> , <code>historyFuel</code> , warnings.
GET <code>/api/suggest</code>	q query	Up to six live-suggestion results with lat, lng, label, state.
GET <code>/api/geocode</code>	q query	Explicit address/coordinate results.
POST <code>/api/routes</code>	points, profile and options	Normalised routes plus provider note.
POST <code>/api/access</code>	items, profile and options	Directional access results keyed by submitted item IDs.
POST <code>/api/stop-path</code>	projection, pump, profile and options	Road connector coordinates for display.

There is no API to save a trip remotely, write a price, edit a token, send a driver message, solve the optimiser, authenticate an operator or synchronise browser history between users. CSV/XLSX parsing

happens locally in the browser; the file is not automatically uploaded to the server or Google Drive.

7.3 Configuration and source request schemas

A public master object contains id, name, gid and sheetId. A public client contains id, name and connections. available has masters, messageMasters and clients booleans. privateMaster reflects credential configuration; truckRouting reflects whether ORS_API_KEY is present. Neither flag guarantees upstream credentials are currently valid.

A source request with masterId must match an active row of the appropriate registry. Otherwise it parses the supplied URL/ID and optional GID as a custom source, with label or Custom Google Sheet as its displayed name. The same source loader is used for fuel and messages, selecting different parsers. Its cache key includes type, spreadsheet ID, GID and label and lasts 30 seconds.

The custom-source path is not restricted to master-listed workbooks. It can read any supported workbook accessible to the configured service account, or an eligible public fallback without that credential. This is intentional flexibility, and also why sharing scope should be limited to workbooks the app is meant to serve.

7.4 Routing request schemas

points is an ordered array of objects with lat and lng. The server accepts 2-25 points in its broad US rectangle. profile=hgv selects ORS; other values use driving. options may include height, width, weight, length, hazmat and avoidTolls. The route response contains routes plus note; each route includes id, name, summary, distance, duration, coordinates, profile and provider.

access items each contain id, projection {lat,lng} and pump {lat,lng}. The server permits 1-20 items, validates both coordinates and the 10.1-mile geographic separation guard. Each returned result provides the requested ID and either confirmed inMiles/outMiles/seconds or a confirmed unreachable outcome/reason. Request-level failure is distinct from a per-stop exclusion. The browser requires a result for every requested item before treating access preparation as complete.

stop-path accepts a single projection and pump with the same profile/options. It returns a round-trip connector geometry. It is a map-display request; recalculation of the selected plan does not occur when the connector is opened.

7.5 Generic outbound fetch helper

fetchData defaults to a 25-second timeout and three attempts. It rejects redirects. It retries 429 and 5xx responses, using a numeric Retry-After delay capped at five seconds or an exponential fallback; network failures have short increasing waits. Responses are read as text, limited to 15 million characters after receipt, then parsed as JSON where expected. This post-read limit is not a streaming memory cap.

The cached helper coalesces concurrent identical requests and applies fixed expiry after successful completion. It caps cache entries at 1,500 and evicts older insertion entries. A failed pending call does not become a successful cached value. Queues and caches are ordinary module memory and therefore specific to a warm function instance.

Source: server.js:1-111, 310-370, 557-689, 694-1013; workspace.js:206-240.

7.6 Exact external destinations in this snapshot

These are source-configured destinations/templates, not a record of production calls made during this review. Placeholder braces describe runtime substitution. Server calls keep authentication headers private; browser map/font calls are ordinary requests visible in the Network panel.

Destination / template	Caller and purpose
https://oauth2.googleapis.com/token	Server: signed service-account JWT exchange.
https://sheets.googleapis.com/v4/spreadsheets/{id}?fields=sheets.properties	Server: resolve GID to current sheet title.
https://sheets.googleapis.com/v4/spreadsheets/{id}/values/{encoded-range}?valueRenderOption=UNFORMATTED_VALUE	Server: private tab values.
https://docs.google.com/spreadsheets/d/{id}/gviz/tq?tqx=out:csv&gid={gid}	Server: public fuel/message fallback only.
https://api.samsara.com	Server: inventory, stats, fuel-energy and history endpoints from Chapter 6.
https://photon.komoot.io/api/	Server: q, countrycode=US, lang=en, limit=6 suggestions; PHOTON_URL can replace base.
https://nominatim.openstreetmap.org/search	Server: q, format=jsonv2, countrycodes=us, addressdetails=1, limit=6; base configurable.
https://router.project-osrm.org/route/v1/driving/{coordinates}	Server: ordinary route and displayed stop connectors; base configurable.
https://router.project-osrm.org/table/v1/driving/{coordinates}	Server: directional distance/duration matrix; base configurable.
https://api.openrouteservice.org/v2/directions/driving-hgv/gejson	Server: optional truck directions and access; base configurable.
https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png	Browser: street raster map.
https://basemap.nationalmap.gov/arcgis/rest/services/USGSImageryOnly/MapServer/tile/{z}/{y}/{x}	Browser: satellite/aerial raster imagery.
https://tiles.openfreemap.org/styles/liberty	Browser: primary Hybrid reference style; its retained source/glyph/sprite URLs drive further requests.
https://services.arcgisonline.com/ArcGIS/rest/services/Reference/World_Transportation/MapServer/tile/{z}/{y}/{x}	Browser: fallback reference roads.
https://services.arcgisonline.com/ArcGIS/rest/services/Reference/World_Boundaries_and_Places/MapServer/tile/{z}/{y}/{x}	Browser: fallback boundaries/place labels.
https://fonts.googleapis.com and returned font hosts	Browser: DM Sans and Manrope font CSS/files.

Map attribution links point to USGS, OpenFreeMap, OpenMapTiles, OpenStreetMap and fallback data contributors. They are attribution/navigation links, not extra fuel-price providers. No Google Maps, WhatsApp send or alternative telematics endpoint is hidden behind these UI controls.

8. Planner page and result workflow

8.1 Layout and pinned actions

On desktop, Planner has an independently scrollable input column on the left and a large map/results column on the right. Find routes stays at the bottom of the left panel so it can be used from any input position. Fuelio Hunt is the pinned action beneath the map. Route alternatives, progress and results can be reached by scrolling the right column; they do not consume a permanently pinned strip that squeezes the map.

The latest stylesheet enlarges the desktop map to nearly the available workspace height, with minimum-height adjustments for short windows. Narrow screens stack the areas and use viewport-relative map/input heights. A ResizeObserver calls Leaflet invalidateSize so panel or viewport changes do not leave an incorrectly sized map canvas. Scroll behavior is controlled by the view/layout CSS, not by changing route geometry.

8.2 Inputs and vehicle selection

The truck control supports an active connected vehicle or manual unit text. A compact pane shows the selected truck, speed, fuel percent and MPG. The long original VIN/engine/odometer card and separate Manual Planning dialog are absent from the updated page. Manual operation is still possible through the editable fields; removing the dialog did not remove that capability.

Origin, destination and separate ordered waypoint rows share typeahead, explicit search, coordinates, outside-click dismissal and keyboard behavior. The visible waypoint order is the actual route order. The former buffer-distance explanatory line is no longer shown. Fuel rules remain editable, with fresh maximum stops set to three. Save Trip is hidden while its code remains retained.

Truck selection can apply live origin, fuel and MPG unless a manual override protects them. New API data do not automatically write changes back to Samsara. A selected truck's geographic movement is reflected on the map and can make a route stale if the unprotected origin changes.

8.3 Planner map before and after route finding

Before there are routes or plans, the map can show all active client vehicles as UFO icons and the loaded stop markers. Vehicles whose names match the case-insensitive deactivated pattern are filtered out. Vehicle marker hover/click shows its Fleet-style information; a stop marker uses the same stop details as Fuel Stops. Clicking a vehicle here is informational and does not silently change the selected planning truck.

Once routes or plans exist, the Planner vehicle layer keeps only the selected truck. A manual trip with no selected live vehicle has no live-vehicle marker in that focused state. Other vehicles remain available in Fleet; the raw fleet data are not deleted. Clearing the trip restores the initial multi-vehicle view.

Routes are drawn in a repeating colour palette. Selected routes are stronger and more opaque; nonselected routes are faint. Route-line clicks toggle route selection. Origin/destination markers derive from the first route's endpoints. The map does not create an independent set of waypoint markers for every intermediate control. Stop markers after route finding use the selected-route corridor/candidate view and deduplicate physical stop IDs for display; the solver still retains occurrence-specific IDs for repeated passes.

The retained Map click mode control can be None, Origin, Destination or Waypoint. In Planner, clicking empty map space in a placement mode writes latitude/longitude rounded to six decimals into that endpoint or adds a waypoint, with the same 23-waypoint cap. Origin placement becomes a manual origin override and marks the route stale. In None mode, an empty click just closes open information. This placement control is not part of the 25-field persisted form whitelist.

8.4 Find routes and Fuelio Hunt

Find routes resolves current locations, requests road alternatives, prepares their geometry, selects the returned routes and displays distance/duration summaries. Fuelio Hunt captures current rules and fuel data, confirms road access and runs each selected route in a worker. It sorts completed results by purchase cost. Part I explains its exact state model, distances and warnings.

A change to truck/client, endpoint, waypoint, profile or routing options requires Find routes again. A change to gallons, MPG, capacity, buffer, filling strategy, stop limit, ending target, corridor or arrival-window rules only needs Fuelio Hunt. Route-staleness messages and disabled/available controls are driven by the recorded trip stamp. The application never interprets a theme change as a calculation change.

8.5 Result cards and actions

Each successful card identifies the route and truck, source master, fill mode, timestamp, total purchase spend, total miles including stop access, gallons to buy, number of stops, detour miles and ending gallons. Stops display identity, city/state, highway/exit, route mile, arrive-buy-leave values, cost and four-decimal price. The lowest displayed successful cost receives the LOWEST FUEL SPEND badge.

Warnings report relaxed strict-window rescue or excluded stops without confirmed road access. Infeasible Half Tank results offer Use Full Tank; this changes the form strategy for the user to re-run, rather than presenting a half-labelled full plan. Zero-stop results explicitly state that current starting fuel covers the trip and reserve.

The expandable fuel profile is an SVG chart with consumption declines, vertical refuelling increases and reserve reference. The underlying balance is displayed in text. Export CSV and Export JSON first replay the plan's arithmetic. Clicking a stop focuses it and can draw a dashed access connector. Driver message opens the same editable composer used on Messages, initialised from that plan.

8.6 Reset behavior

Clear trip asks before removing the current trip. It clears truck/origin/destination/waypoints, routes, plans, route stamp, result cards and connector layer, then redraws the map. It preserves fuel data, saved routes and history. Other fuel-rule form values are not all reset to factory defaults. This is distinct from Fuel Stops Clear, message Clear and browser backup restoration.

The Clear trip handler does not itself explicitly abort an active solve worker. If clearing while a hunt is still running, use the existing Cancel control first; that reset-during-solve interaction should receive a regression check before changing its behavior. The completed-trip reset behavior is covered by the current page tests.

Source: index.html; workspace.js:285-368, 534-684, 867-1003, 1038-1647, 1953-2181; styles.css later planner override blocks.

9. Fuel Stops page: complete interaction contract

9.1 Shared records and list/map layout

Fuel Stops places a searchable list/table on the left and the interactive map on the right on desktop. Both are built from the same normalised loaded records and source metadata. On smaller screens the areas stack. The active master and loaded stop count remain visible as source context; they are not recomputed from a different dataset for the map.

Existing source loading remains: select an active master, load it, use a custom Google Sheet URL/GID, or upload CSV/XLSX. Changing basemap, theme, filter or sort does not modify prices or load a different master. A foreground source load only replaces the actual dataset when parsing/fetching succeeds. A background refresh preserves previous values on failure.

9.2 Live search and ranking

Search responds to typing without Enter. It searches stop name/number, city, state, location, highway, exit and price, including formatted price text. The helper normalises case and common punctuation and expands Mt/Mount, St/Saint, N/North, S/South, E/East, W/West and Mc/Mac variations. Expansion supplements normal text matching rather than replacing it.

There is no special search rewrite from California, Cal or Cali to CA. An exact state-name occurrence in ordinary text can still match normally; the state dropdown remains the explicit state-code filter. Import state normalization is separate and still maps an exact full state name to its abbreviation.

With original/master order selected, an exact numeric stop-number match ranks above an incidental match such as US-200. Other results keep stable master order. Explicit price sorting takes precedence over this number priority. Numeric identity handling and leading-zero stripping happen in the resolver/ranking helpers; a search string still needs to match the searched text to enter the result set.

9.3 State filter

The State dropdown is generated from states present in the currently loaded data, sorted, with All as default. It is not a fixed list of every US state. State filters the table. Its effect on the map depends on whether one or more highways are selected, as specified below. Replacing the source rebuilds the state choices.

9.4 Multi-select highways

The highway menu contains recognised highway tokens with pump/stop counts from the complete loaded dataset. Its own text search filters the menu's options. Multiple selected highways are ORed: a stop matching I-40 or I-10 is accepted. This highway group is then ANDed with the stop search and state filter.

The small x in the menu clears only highway-search text. It preserves selected highways. Clear highways removes selections. Clicking outside closes the menu without clearing its selections. Counts are not dynamically recalculated from every current text/state filter; they describe the loaded source's highway-token distribution.

9.5 The map/filter rule

Highway selection	Table rows	Map markers
No selected highways	Search AND State	All loaded stops, regardless of Search or State.
One or more highways	Search AND State AND (Highway 1 OR Highway 2 ...)	The same filtered stop set as the table.

This behavior is implemented in stopMapRows and the workspace's view-specific map selection. It deliberately lets the map remain a full-master overview during ordinary text/state browsing. It must not be replaced by a generic rule that every filter always filters the map.

Price sorting changes table order, not which markers exist. Viewer filters are not restrictions on Fuelio Hunt's loaded fuel source. The hidden older routeOnly checkbox remains for compatibility, but it is not the current viewer rule or an automatic nearby-stops planner feature.

9.6 Price cycle and export

Clicking the Price header cycles original master order, highest to lowest, lowest to highest, then original again. The cycle applies to the current filtered rows. Equal-price rows keep their source order. The master-order mode is also where exact stop-number search priority applies.

Export downloads the current visible/filtered rows, rather than every loaded row unconditionally. The filename is fuelio-fuel-stops.csv. It includes pump_name, price_per_gallon, latitude, longitude, city, state, brand, store_number, highway and exit. These compatibility headers are accepted on re-import even though the UI calls the entities stops.

9.7 List-to-map and map-to-list selection

Clicking a table row highlights it, flies to the same stop at at least zoom 8 and opens the popup. It does not automatically scroll the table. Selection is stored by stop ID. Clicking a marker selects the same record and highlights its table row.

If Highway is off and a marker was hidden from the list by Search/State, marker selection clears those list filters so the chosen stop becomes visible. If it was already visible, filters are preserved. Highway selections cannot produce a visible marker outside their own filtered set, so that special clearing is for the highway-off full-map case. Clicking empty map space closes popups/tooltips without resetting search, state, highway selection or price ordering.

9.8 Stop details

Hover and click expose stop name, location, price, highway, exit and latitude/longitude. Price is formatted to four decimal places and coordinates to five. Missing highway/exit fields stay empty; the map does not reverse-engineer them. Text is HTML-escaped before insertion into popup markup.

9.9 Clear and Fit

Clear empties stop search, resets State to All, removes all highway selections, clears highway-search text, resets price sort to master order, clears selected/open stop state, restores the full list and marker set, and fits the map. It does not change master, theme or basemap mode. The source remains the same.

Fit uses the currently displayed marker set for Fuel Stops. With no markers it safely returns to the default US view. Dragging, wheel zoom, plus/minus, fullscreen and the imperial scale remain available. Map/Satellite changes never alter the record order or selected price values.

Source: view-model.js; workspace.js:763-1037, 2058-2144; data.js; styles.css stop-view layout.

10. Fleet page: complete interaction contract

10.1 Fleet selection and visibility

Fleet uses the current client and its combined connections. Active vehicles appear in a left-side list with a corresponding map on the right. The search matches name, VIN and location and filters both the list and vehicle markers. The name filter excludes any vehicle matching /deactivat/i. It does not inspect a Samsara status field and does not match every possible spelling such as deactivate.

This is a presentation filter. It does not delete vehicles from Samsara or change their inventory records. A truck that lacks valid GPS can remain useful as a list record even when it cannot be plotted. A missing reading is represented as unavailable, not silently zero.

10.2 List data and detail hydration

Cards show the truck, location and telemetry such as fuel percent, MPG, speed, odometer and GPS timing where available. Percent and MPH are rounded for display; MPG uses two decimals. Basic snapshot values may load before report-derived MPG. A 60-second fleet refresh and the separate detail-hydration mechanism update those displays without requiring a full page reload.

Fuel can come from the current snapshot or a labelled latest-history fallback. The source timestamp remains important. A truck's 30-day MPG is not recalculated from its current speed. The app can continue to display last known values with warnings during a partial connection failure.

10.3 Vehicle icon and information

Fleet uses the bundled rocket SVG in place of the earlier arrow. Heading is normalised into 0-360 degrees; missing/invalid heading uses zero for visual orientation. The icon turns to follow the direction reading, but it does not extrapolate a future position or animate a predicted path between GPS samples. Planner uses the separately bundled UFO icon.

Hover and click show truck name, location, rounded speed in MPH, fuel percentage and MPG. Clicking a list card focuses its truck on the map at at least zoom 10 and opens its information. Map marker selection updates focus and the corresponding list state. New telemetry can update the displayed popup/tooltip while preserving the selected truck.

10.4 Fleet-only control differences

There is no Plan trip button in the updated Fleet list. Price Heat is hidden because prices are irrelevant to the fleet vehicle layer. The Fleet map does not show fuel-source name or pump-count captions. Basemap switching, zoom, pan, fit, fullscreen and imperial scale remain shared map facilities. Fit uses the filtered vehicles with usable coordinates.

Selecting another client clears the current truck selection, existing client vehicle array, origin and starting-fuel fields before loading the new fleet. It increments request guards to discard old responses. The fuel-price master remains an independent selection, as documented in Chapter 3.

Source: view-model.js activeVehicles, mergeVehicleDetail and vehicleIconHTML; workspace.js:449-684, 915-984, 1953-2043.

11. Messages: one editor in two places

11.1 Standalone and Planner parity

Messages and the Planner's Driver message dialog instantiate the same MessageEditor component. Both support driver name, optional team-driver name, editable route, selected message master, manual stop numbers/names, alternative stops, small editable stop details, CSV/XLSX loading, Google Sheet loading, template download, Sheet instructions, separate primary/team previews, copy buttons and clear controls.

There is no send/WhatsApp/email action. Copy places text on the clipboard, with a browser fallback when the Clipboard API is unavailable. It does not contact a driver or record a delivery acknowledgement. The old Filling instruction field and generic Please follow the planned fuel quantities suffix are absent from the current shared composer.

11.2 Manual identifiers and resolution

A user can type 245, 211, #245, a stop name or a full suggested stop line. Datalist suggestions help find a matching record but the field remains editable. Exact ID/name/detail matches are considered, then normalised number matching. Multiple materially different records with the same number are treated as ambiguous rather than choosing a different brand silently.

The resolver compares a descriptive tuple containing brand, number, location, highway and exit so identical catalog duplicates do not necessarily make an identifier ambiguous. The user can select a more specific suggestion or enter/edit details manually. A manual stop needs an identity and location for a useful message; missing highway/exit can remain blank.

11.3 Stop pairs

Each pair has a primary stop A and optional alternative B. A used pair requires its first stop. Alternatives remain blank until the user supplies them. The user can add or remove pairs and edit every stop field, including planner-proposed stops. A second alternative is separated by OR on its own line. The star count represents pair order, not an arithmetic quantity or a Markdown bullet.

The message formatter rejects repeated stop identities. Its duplicate key prefers record ID; manually created records can have pair-specific IDs, so semantic duplication is not completely prevented for independently typed manual entries. There are no dedicated drag/reorder controls for message pairs in this snapshot; order can be changed through editing/removing/adding entries. Waypoint reordering is a separate implemented feature on Planner.

11.4 Planner autofill

Opening a plan message first verifies the selected plan. It fills primary stops from that plan and leaves every alternative blank. Catalog enrichment first requires a unique numeric catalog match, then checks that its brand equals the planned stop's brand before using its details. An ambiguous mixed-brand catalog keeps the original planned record instead of arbitrarily enriching it. This prevents a Pilot #245 from silently becoming Love's #245. The operator may change the proposed station or any descriptive text afterward.

The route field extracts recognised highway references from the route summary. It recognises I, US, SR and selected state-prefix forms including CA, TX, AZ, NM, OK, IL, IN, OH, PA, NY, NJ and FL. It is not a full highway database parser. If no recognisable highway exists, the field stays blank for manual entry. The route label is not forced to become a long coordinate pair string.

Plan-specific drafts are separate from the standalone Messages draft and from other plan IDs. Editing one does not silently change the other's driver names or alternatives. Plan drafts live in session memory; the standalone message draft is saved in IndexedDB. A source/catalog switch resets stop selections rather than leaving old catalog identities incorrectly linked, while retaining the editable general driver/route fields.

11.5 Primary-driver template

The example below shows the current formatter's literal text. The actual source uses at these fuel stops in the closing sentence, reflecting the requested Stop wording. Double spaces after a stop number and literal asterisks are preserved in copied text.

Hey Fid, I have listed down the suggested Love's fuel stations with their nearest city for your consideration.

*245 Oklahoma City, OK (HW: I-44, Exit: 113)

OR

*211 Oklahoma City, OK (HW: I-35, Exit: 120)

**712 Newcastle, OK (HW: I-44, Exit: 107/24th St)

OR

**88 Yukon, OK (HW: , Exit:)

We will appreciate if you fill up the tank at these fuel stops. This Result is based on the route (I-10). If you change your route please let me know.

Drive Safe!

In the actual copied message, the greeting and closing paragraphs are each continuous paragraphs; the line wraps above are only for the PDF page width. If the primary name is blank, the greeting is Hey, rather than inventing a driver. If all resolved stops share a brand, the greeting names it. Mixed brands use generic fuel stations.

11.6 Team-driver template

Hey Vid, We spoke to Fid, I have listed down the suggested Love's fuel stations with their nearest city for filling up your vehicle tank. The company is getting a discount on filling up at these fuel stops.

*245 Oklahoma City, OK (HW: I-44, Exit: 113)

OR

*211 Oklahoma City, OK (HW: I-35, Exit: 120)

**712 Newcastle, OK (HW: I-44, Exit: 107/24th St)

OR

**88 Yukon, OK (HW: , Exit:)

We will appreciate if you fill up the tank at these fuel stops. This Result is based on the route Fid discussed with us (I-10). If you change your route please let me know.

Drive Safe!

Both driver names are required for the team version. The team message is separately generated and editable; it is not just the primary message with the salutation replaced. It references the primary driver in both the opening and route paragraph. The discount sentence is static template wording, not a checked comparison between retail and contract prices.

11.7 Preview editing, persistence and limits

Primary and team previews can each be edited before copying. Changing a structured field or stop can regenerate the template, replacing manual preview edits. Draft snapshots include the current structured entries, source/catalog state and editable preview text. Standalone saves are debounced by about 200 ms once the editor is ready. Clear resets the relevant composer state; it does not delete a master from Google Sheets.

The editor snapshot format is version 2: rows, source, master, pairs, fields and previews. Each pair side stores text, record, manual and ambiguous. Restoration also accepts the earlier `messageDriver/messagePartner/messageRoute` field names and older string-valued pair entries, resolving them against the current catalog. Version-2 manual previews are restored only after structured inputs validate. Invalid/ambiguous stop inputs clear generated previews and disable copying until corrected.

The current template deliberately contains no computed gallons, purchase dollars or ending-fuel quantity. The retained `planMessage` function in `messages.js` can produce the older quantitative text, but the current UI uses `MessageEditor/customMessage` instead. Consequently a Half Tank result still initially says fill up the tank in its suggested prose. An operator can edit that wording; it is not automatically aligned with $Q = C/2$. A zero-stop result has no suggested pair to populate, so the general editor is not a specialised zero-purchase dispatch message.

Source: `message-editor.js`; `messages.js`; `view-model.js` `routeHighways` and `stopNumber`; `workspace.js`:1588-1647.

12. Maps, Hybrid references and Price Heat

12.1 Leaflet base and layer order

The shared map supports dragging, mouse-wheel zoom, plus/minus controls, Fit, fullscreen, imperial distance scale and Map/Satellite switching. The user's last basemap mode is stored in browser state. Switching it removes/adds imagery/reference layers without changing stops, prices, filters, selected records or table sort.

The default street view uses OpenStreetMap tiles. Satellite uses the existing USGS imagery-only base and adds reference layers. Fuel markers, vehicles and route overlays remain separate from the reference-map stack. The map can be usable with only imagery if external reference services fail, but that is a degraded visual state rather than proof that the road labels loaded.

Layer	Intended stacking / role
Satellite imagery	Base raster tiles; max zoom 19, native imagery zoom 16.
Raster reference fallback	Pane z-index 300; road and place labels while vector references are unavailable.
Vector Hybrid references	Pane z-index 350; transparent roads, route references and place symbols.
Route and stop overlays	Above reference panes; ordinary Leaflet overlays around z-index 400.
Vehicle markers / popups	Marker/popup panes above map reference information.

Reference panes do not intercept pointer events, so labels do not block stop/vehicle selection. The source keeps imagery neutral by overriding theme tile filters for the satellite and reference layers.

12.2 Primary Hybrid vector style

hybrid-map.js fetches the OpenFreeMap Liberty style and transforms it into a transparent reference-only style. It keeps symbol layers and road/transport/highway/bridge/tunnel/boundary line layers, removes opaque backgrounds/area fills and discards unused sources. The resulting style overlays the USGS aerial imagery rather than replacing it with a second opaque map.

Place and road text is white with a dark #142331 halo of approximately 1.8 pixels. Major roads use warm #ffd18a lines, local roads light #f2f0e6, dark #27333d casings, subdued rail/boundary colours and optional icons. The intent is readable geography over varied aerial terrain without tinting the imagery itself.

Highway-shield layers are rebuilt as route-reference text with line placement, minimum zoom 6, spacing 260 and a 2-pixel halo. Numeric references can be prefixed I-, US- or SR- based on their network metadata, while already prefixed references remain intact. This does not invent missing highway data or guarantee every road shield at every zoom. It prioritises readable references over an exact copy of a proprietary hybrid basemap.

12.3 Load, readiness and fallbacks

The primary style request has a 12-second abort timeout. Its fetched style can be reused, but a fresh cloned style object is passed to each MapLibre instance. Raster World_Transportation and World_Boundaries_and_Places references are immediately available as fallback while the vector overlay loads.

Once the vector map reaches idle and reports rendered reference features, the raster fallback is removed. If loading errors occur, the plugin is unavailable, or readiness is not established within approximately 16 seconds, the fallback remains. A generation counter and show/hide lifecycle stop late asynchronous work

from resurrecting Hybrid references after switching back to street mode. The local tests cover successful readiness, fallback and cancellation behavior with fake map objects.

Fallback services are external legacy Esri services, not bundled tiles owned by Fuelio. Esri has published a retirement programme for legacy basemaps; maintainers should verify the specific fallback service lifecycle and replace it before its retirement. The primary vector overlay reduces dependence on the old raster labels but does not remove all external availability risk.

Reference: [Esri legacy basemap retirement notice](#). Source: `hybrid-map.js`; `workspace.js`:767-857.

12.4 Price Heat is a price colour scale

Price Heat is a separate toggle, not a fifth page theme and not a density heatmap. It colours stop markers along a green-to-red HSL scale using the minimum and maximum prices of the current map stop set. Equal prices use the middle of the range. A legend shows the corresponding per-gallon range.

Selected/recommended marker treatment can override the normal heat colour so recommendations remain recognisable.

Price Heat stays at the top of relevant stop/planner maps. It is hidden on Fleet. Turning it on does not sort the table, change a master, recompute routes or affect the optimiser. Changing theme must not darken or colour-shift the Hybrid imagery.

12.5 Fit, focus and empty cases

On Fuel Stops, Fit uses currently displayed stop markers, following the special highway-on/off rule. On Fleet it fits filtered vehicles with coordinates. On Planner, selected route bounds take priority, with map padding and a maximum fit zoom; otherwise the loaded stop set can supply bounds. No usable bounds returns to a default view near latitude 38, longitude -97 at zoom 4.

List selection flies to a stop at minimum zoom 8 or a Fleet vehicle at minimum zoom 10. Empty-map clicks close popups/tooltips without clearing filters. The map centre/zoom is not explicitly saved as a full viewport record; basemap mode, heat and selected/filter state are persisted. A later Fit or new route can change the viewport.

Source: `workspace.js`:763-1037; `hybrid-map.js`; `view-model.js`; `styles.css` map pane rules.

13. Themes, branding, responsiveness and accessibility

13.1 Four page themes

Theme	Core palette	Intended visual treatment
Light	Background #f3f5f9; white panels; accent #2457da	Bright neutral workspace.
Dark	Background #111419; panel #1b1f26; accent #e7bc69	Dark charcoal with gold accents, fine gold details and warmer text.
Crimson	Background #1c1018; panel #2a1923; accent #ff7898	Deep red/pink-tinted interface panels and controls.
Icy	Background #0d293d; panel #183d55; accent #86e3ff	Deep cool blue with cyan highlights, visibly distinct from Light.

Theme selection applies to the entire workspace, message dialogs and controls through CSS custom properties and data-theme. It is persisted locally. Satellite imagery/reference layers bypass theme filters; the UI surrounding the map still uses the selected theme. Street tiles retain theme-specific brightness/saturation treatment in Dark, Crimson and Icy.

The Dark update adds restrained gold accents to active navigation, action styling, panel detail and small decorative features. It does not introduce new analytics or change fuel status semantics. Selected route colours and recommended-stop emphasis remain data cues distinct from the decorative palette.

13.2 Branding and vehicle assets

fuelio-brand.png is the supplied 2048 by 682 brand artwork: a blue fuel/drop-style F symbol and Fuelio wordmark. The header uses this artwork with CSS sizing/cropping rather than an invented replacement brand. The older FCO Dispatch Workspace subtitle is not the intended visible header. The inline favicon still uses a simple F graphic, and package/health/storage identifiers remain legacy FCO values.

vehicle-rocket.svg and vehicle-ufo.svg are separate small source assets. Fleet uses the rocket; Planner uses the UFO. Heading rotation is applied in HTML/CSS around the chosen SVG. Missing heading affects orientation only and does not change coordinates, route geometry or any fuel calculation.

13.3 Layout breakpoints

Desktop uses a 100dvh application shell with hidden outer overflow and independently scrolling work areas. Header/source bars occupy their own vertical space. Planner's left column is normally about 334 pixels, expanding to about 365 on very wide screens and narrowing near 305 on intermediate desktop widths. The right map gets most of the remaining width and height.

At widths at or below roughly 760 pixels, the application stacks content and relaxes the fixed outer shell. Planner's map uses about 68dvh with min/max bounds; the input panel has its own capped viewport-relative height. Fleet/Stops use stacked list/map layouts. Dialogs remain bounded by the viewport and scroll internally. Some intermediate Fleet/Stops widths preserve relatively wide minimum list widths, so very narrow desktop windows should be checked visually.

styles.css includes older base rules followed by page-update and Hybrid-update overrides. Later applicable rules win. Editing only an early selector can therefore appear to do nothing. Maintainers should inspect the effective selector cascade for the view and breakpoint rather than deleting retained rules indiscriminately.

13.4 Accessibility features and limits

The source uses labelled form fields, semantic buttons, dialog controls, visible focus outlines, screen-reader-only text, map/chart aria labels and a keyboard-operable combobox/listbox pattern. Address suggestions support ArrowUp/ArrowDown, Enter and Escape, with aria-expanded and active-descendant updates. Text inserted from source records is escaped or assigned through `textContent` in the relevant renderers.

CSS includes reduced-motion handling for transitions, but that does not necessarily suppress every Leaflet fly-to animation. Tests exercise DOM behavior rather than a screen-reader audit, full contrast certification, touch-device survey or every browser's fullscreen/clipboard support. Those capabilities should be checked in the team's actual browsers before claiming complete accessibility conformance.

Source: `index.html`; `styles.css`; `place-search.js`; `view-model.js`; `fuelio-brand.png` and both vehicle SVGs.

14. Persistence, refresh and cache reference

14.1 Browser preferences

localStorage key `fco.workspace.v1` contains form values, theme, selected client/master, source preferences, manual overrides, mapMode, Price Heat and stop-view state. Stop-view state includes query, state, selected highways, highway-search text, sort mode and selected stop ID. Form persistence uses a whitelist of 25 IDs; the complete list appears in the HTML/control appendix and covers trip, fuel rules and routing restrictions.

Local preferences belong to this browser profile and origin. A different browser, private window, device or deployment domain does not automatically share them. The server does not store each operator's form. Clearing browser storage removes these preferences. A storage failure is reported through a toast rather than preventing a trip calculation.

14.2 IndexedDB data

Data key	Stored value / purpose
fuel	Last loaded normalised source and rows.
saved	Retained saved-route definitions for the hidden library.
history	Up to 30 successful plan snapshots.
loads	Imported/prepared/batch-processed historical load records.
access	Serialised browser road-access cache entries.
messages	Standalone message editor catalog, structured draft and previews.

The database is `fco-workspace`, version 1, with object store data. Read errors return null. Write failures report unavailable/full storage and let the live page continue. This is convenience persistence, not a durable shared operational database, encrypted vault or backup service.

Current route cards, selected routes, fleet samples, activity log and plan-specific message drafts are primarily in memory. Successful plans are copied into history after a complete selected-route solve cycle. An error after an earlier route completed can leave a visible result that was not yet added by the final history step. Browser history does not automatically update when source prices or truck location change.

14.3 Backup and restore

Export backup downloads `fco-workspace-backup.json` with `version=1`, settings, fuel, saved, history, loads and the standalone message snapshot. It does not include service-account/Samsara secrets, server caches, the access cache, active route-search state, transient fleet data, the activity log or all per-plan message drafts.

Restore parses JSON, checks supported version and required saved/history arrays, and replays every historical plan before asking to overwrite the current browser workspace. It writes saved/history/loads, conditionally writes supplied fuel/messages, replaces local preferences and reloads the page. Missing fuel/messages fields are not unconditional deletion instructions. It is a browser-workspace restoration, not a GitHub rollback or Google Sheet restoration.

Only the expected plan replay is applied to history; the backup importer is not a comprehensive schema validator for every possible nested field. Backup files can contain locations, client names, prices and messages even though they contain no configured provider tokens. Treat them as operational project data.

14.4 Complete timing/cache table

Item	Duration / cap	Scope
Browser scheduled refresh	Every 60 seconds, plus visible-tab return	Only while the page is running; overlapping refresh cycles prevented.
Browser API helper timeout	180 seconds by default	Used only without a supplied signal; a caller's signal replaces it.
Google JWT lifetime	One hour in signed claim	New access token exchanged server-side.
Google access token cache	50 minutes	Warm server process.
Spreadsheet metadata	10 minutes	Per workbook in server cache.
Master registry	55 seconds	Combined registry result.
Fuel/message source data	30 seconds	Type + sheet ID + GID + source label.
Samsara inventory	10 minutes	Per token.
Samsara merged token fleet	50 seconds	Per token.
Server vehicle details	55 seconds	Client + vehicle.
Browser complete details	Up to 15 minutes	Client + vehicle.
Browser missing/forced details	About 55 seconds	Shorter retry window.
Address suggestions/geocodes	One day	Lower-cased query, separate cache keys.
Directions	Six hours	Points + profile + options.
Road access	One day	Server request cache and browser geometric cache.
Browser access entries	3,000 maximum	Persisted in IndexedDB.
General server cache entries	1,500 maximum	Warm process; insertion-order eviction.
Retained stat token maps	100 maximum	Warm process only.
Successful history	30 plans	IndexedDB.
Activity log	200 entries	Current page memory.
Message autosave	About 200 ms debounce	Standalone draft after initialisation.

14.5 What live every minute really means

Refresh first reloads configuration, then requests fleet and fuel-source updates with Promise.allSettled. Upload-based sources have no external spreadsheet to poll. Message catalog selections are loaded through their own controls; the 60-second loop does not reload every message master or all client fleets. Changing the registry can update available choices without continuously downloading every linked workbook.

A browser background tab can be throttled or suspended, a device can sleep, caches can retain samples, and upstream measurements can already be old. Thus 60 seconds is the intended refresh cadence, not a

guarantee of data age. When a refresh fails, the UI reports retained last values rather than clearing useful data. No Vercel cron, websocket feed or closed-browser scheduler is configured.

The former 300-second fleet failure was a server execution-timeout symptom. A good config response alone established that configuration could be read, not that every Samsara page/report completed. The present snapshot contains bounded fleet/report deadlines and partial-connection handling specifically relevant to that failure mode.

Source: workspace.js:45-187, 603-661, 1204-1281, 1358-1415, 2201-2361; server.js cache/deadline helpers.

15. Activity, saved routes and hidden load tools

15.1 Activity

Activity renders the in-memory dispatch log and successful plan history. Logs include local display times, message text and type such as info, success or warning. New entries are bounded to 200. Clear log removes those in-memory entries. There is no server-side audit table, automatic message-delivery record or multi-user activity synchronisation.

History stores full successful plan snapshots, including route, source, rules, purchases and timestamp. Opening a history result replays its arithmetic and restores its result for inspection. It does not make that old plan current against today's prices. Its route stamp is cleared, so a new calculation requires the current trip's route workflow rather than treating historical geometry as automatically fresh.

15.2 Retained hidden workflows

The hidden Routes & loads page still contains saved-lane cards, Save Trip, route-library export, load import, cluster-radius preparation, batch planning and load-result export code. The navigation item, page routing and relevant controls are hidden. This is reversible presentation gating; it is not code deletion or a change to optimizer.js, geo.js or batch.js.

The saved route shape contains its identity/name, origin/destination, waypoints/points and source metadata. Saving uses the current trip's resolved coordinates. Export saved routes writes fco-saved-routes.json from state.saved; it does not export the bundled library as if the user had created it. The ordinary Planner does not automatically add the hidden library's lane alternatives to every Find routes call.

15.3 Complete bundled route anchors

Stored place label	Latitude	Longitude
Fresno, CA	36.7378	-119.7871
Tulsa, OK	36.1539	-95.9928
Cheshire, CT	41.5084	-72.9106
Baltimore, MD	39.0458	-76.6413
Chicago, IL	41.8781	-87.6298
Sacramento, CA	38.5816	-121.4944
Los Angeles, CA	34.0522	-118.2437
Atlanta, GA	33.749	-84.388

pur-1 and pur-2 connect Fresno/Cheshire in opposite directions, both via Tulsa. pur-3/pur-4 connect Fresno/Baltimore; pur-5/pur-6 Fresno/Chicago; pur-7/pur-8 Sacramento/Chicago; pur-9/pur-10 Los Angeles/Chicago; pur-11/pur-12 Fresno/Atlanta. Those other ten definitions have empty waypoint arrays. Every entry is labelled PurFCO route library. These are supplied anchor coordinates, not verified facility entrances.

15.4 Load template and alias map

load-template.csv is a header-only file with load_id, pickup_city, pickup_latitude, pickup_longitude, delivery_city, delivery_latitude, delivery_longitude, starting_fuel, tank_capacity, mpg and ending_fuel. Header normalisation is the same shared lower-case/alphanumeric conversion; camel-case versions such as LoadID normalise equivalently.

Load field	Accepted normalised headers
ID	loadid, tripid, id
Pickup label	pickupcity, origincity, pickup
Pickup latitude	pickuplatitude, originlatitude, pickuplat
Pickup longitude	pickuplongitude, originlongitude, pickuplng, pickuplon
Delivery label	deliverycity, destinationcity, delivery
Delivery latitude	deliverylatitude, destinationlatitude, deliverylat
Delivery longitude	deliverylongitude, destinationlongitude, deliverylng, deliverylon
Starting gallons	startingfuel
Capacity	tankcapacity
MPG	mpg
Destination target	endingfuel

The imported load object has id, pickup, delivery, overrides, status and matches. Preparation adds cluster IDs and eligible directional lanes based on actual endpoint distances. Batch processing can attach a plan and updated status. The default radius is 50 miles. Clusters are connected-component summaries and never replace actual endpoint eligibility, as explained in Part I.

Load export includes Load ID, Pickup, Delivery, Pickup cluster, Delivery cluster, Lane, Status, Fuel cost, Gallons, Stops and Ending fuel. The current Lane column uses the first eligible match; it can differ from the lane behind the winning plan. Old lane/review functions also use retained hidden waypoint paths, which should be checked when restoring their visible UI.

Source: route-library.json; batch.js; load-template.csv; workspace.js:1648-1882, 2182-2241.

16. Build and deployment operations

16.1 Repository and scripts

Keep the source files together at the repository root used by Vercel, with package.json, server.js, index.html and build.js at that root. package.json declares type=module, private=true and Node 22.x. Its scripts are:

Script	Command / purpose
npm run build	node build.js; prepare public assets.
npm start	node --env-file-if-exists=.env server.js; local server with optional .env.
npm test	node --test tests.mjs; full supplied suite.

Use npm ci to install the lockfile's dependency set before a fresh build. For local use, run the build then npm start and open the local server, normally port 3000. Opening index.html directly as a file is not the supported way to exercise same-origin APIs, modules, workers and runtime assets.

The source archive does not contain a .gitignore file. Keep real credentials, local .env, generated dependencies and unneeded build artifacts out of commits through the repository's actual ignore/secret-management setup. This documentation did not add an ignore file or alter the repository.

16.2 Public build contents

build.js copies 21 explicit project assets: index.html, styles.css, workspace.js, optimizer.js, geo.js, data.js, messages.js, batch.js, worker.js, route-library.json, california.json, fuel-template.csv, load-template.csv, message-template.csv, message-editor.js, view-model.js, place-search.js, hybrid-map.js, both vehicle SVGs and fuelio-brand.png.

It then copies ExcelJS's minified browser file, Leaflet's distribution, MapLibre GL JS and CSS, and the Leaflet/MapLibre bridge into public/vendor. It creates THIRD-PARTY-NOTICES.txt from the Leaflet, MapLibre GL, bridge and ExcelJS license files. It checks that the copied HTML references workspace.js.

Server source, environment setup, tests, package files and documentation are not on the project's public-copy whitelist. The build does not minify or bundle the project's own modules; imports remain browser ES-module requests. It creates/overwrites destinations but does not clear an existing public directory first, so local stale generated files can remain unless the build directory is clean. Vercel's fresh deployment build normally avoids relying on a manually maintained public folder.

16.3 Vercel configuration

vercel.json declares framework=express and buildCommand= npm run build. The API cache header is private, no-store. Site-wide headers add nosniff, strict-origin-when-cross-origin referrer policy and SAMEORIGIN framing. There is no cron entry, database binding or maxDuration override in this file.

server.js exports the Express application. For direct local execution outside Vercel it listens on PORT or 3000; on Vercel it relies on the framework deployment. Vercel's documented Express integration serves static files from public through its static/CDN path and treats the Express application as a function. This is why build.js is essential and why replacing only index.html cannot repair server-side telemetry logic.

Reference: [Express on Vercel](#). Source: vercel.json; server.js final handlers; build.js.

16.4 All environment variables read by the application

Variable	Default / purpose
GOOGLE_SERVICE_ACCOUNT_JSON	Complete private Google credential JSON; no default credential.
MASTER_SHEET_ID	Default 1VpTN7OY40d1yVFLvtBu2otIbUjAjqm6DVy71FDsBeeY.
FUEL_REGISTRY_GID	0.
MESSAGE_REGISTRY_GID	970811620.
SAMSARA_REGISTRY_GID	1100093231.
SAMSARA_TOKEN	Optional direct token fallback if no clients are found; comma/newline-separated.
SAMSARA_CLIENT_NAME	Connected fleet for direct fallback.
ORS_API_KEY	Optional truck-routing credential; presence enables HGV UI capability.
OSRM_URL	https://router.project-osrm.org ; ordinary route/matrix provider base.
ORS_URL	https://api.openrouteservice.org ; HGV provider base.
PHOTON_URL	https://photon.komoot.io ; live-suggestion provider base.
NOMINATIM_URL	https://nominatim.openstreetmap.org ; explicit geocoder base.
GEOCODER_USER_AGENT	App-identifying User-Agent for explicit geocoding; configure a real deployment/contact string.
PORT	3000 for direct local execution.
VERCEL	Platform-provided marker used to avoid starting a local listener.

PHOTON_URL exists in server.js but is absent from the older ENVIRONMENT.txt list. There are no required NEXT_PUBLIC_ or VITE_ variables. Secrets must remain server-side. OSRM_URL and ORS_URL should be base URLs, not complete individual route request URLs; the server appends its known API paths.

16.5 Credential replacement procedure

For a rotated Google key, replace GOOGLE_SERVICE_ACCOUNT_JSON with the complete new JSON object, without wrapping it in an extra quoted JSON string. Preserve the private_key escapes and ensure client_email matches the service account that has workbook access. Save the variable for the intended Vercel environment and create a new deployment so that environment value is used.

If a different service account was created, share the Master and required private linked workbooks with its new email. If /api/config reports no errors and all availability flags true, registry access is working for that request; diagnose fleet failures at /api/fleet and optional report failures at /api/vehicle-detail rather than repeatedly changing a working Google key.

Samsara token changes belong in the private client row or optional server fallback. They are picked up after configuration cache expiry and refresh. A token's validity, endpoint permissions and inventory coverage are separate from its existence in a Sheet cell. A Google service-account key cannot replace a Samsara bearer token.

16.6 Hosting and external-service limits

The reviewed source is configured for a Vercel Express deployment and does not require a paid solver or database. That does not guarantee unlimited free traffic from all dependencies. At the documentation date, Vercel's published limits list a 300-second Hobby function maximum and a 4.5 MB function request/response payload limit. The app's own 2 MB JSON parser and per-call deadlines can be lower. Do not solve timeouts by assuming a larger platform budget is available.

Reference: [Vercel function limits](#), checked 13 September 2026.

Google's published Sheets limits list 300 read requests per minute per project and 60 per minute per user/project; service-account calls count as one account. Standard Sheets API use is described as having no additional cost, with announced changes concerning over-quota billing later in 2026. Check the actual project's current quotas and billing terms before scaling; a free service-account identity does not remove API quotas.

Reference: [Google Sheets usage limits](#), checked 13 September 2026.

Public geocoders, route servers, imagery/vector services and Samsara have their own availability, terms and account limits. The code's process-local pacing and caches reduce repeated calls but cannot enforce a single global limit across many Vercel instances. The default public Nominatim service has strict application-wide restrictions and forbids autocomplete; Fuelio uses Photon for typing suggestions and Nominatim only for explicit lookup. No external quota is represented as unlimited in this manual.

16.7 Updating and rolling back

Before replacing files, preserve the current repository revision and use the existing test/build commands. Interface changes commonly involve index.html, styles.css, workspace.js and a focused helper. Algorithm or provider changes have different impact areas. Ensure every new browser asset is added to build.js; otherwise it may work locally from source while missing from the deployed public output.

GitHub-connected Vercel can build a new deployment from the updated commit. Inspect that deployment's build/runtime logs and verify the deployed revision before diagnosing browser data. Rolling back a deployment restores code, not necessarily changed Google Sheets data or browser-stored preferences. Browser backup and repository rollback are separate recovery mechanisms.

Source: package.json; package-lock.json; build.js; vercel.json; ENVIRONMENT.txt; server.js.

17. Diagnostics, privacy boundaries and failure recovery

17.1 Diagnostic ladder

Symptom / observation	Meaning and next useful evidence
HTML/CSS fails to load	Check build output/public assets and deployment revision before API credentials.
/api/health works	Express responds; this does not test Google, Samsara, routes or map tiles.
/api/config has errors	Inspect which available flag is false; verify that registry tab's GID, share and credential.
Config succeeds but fleet fails	Focus on Samsara connection warnings/status, inventory access, pagination and timeouts.
Some trucks load	Inspect partial and connection counts; one client token may have failed.
Locations load but fuel is blank	Snapshot may not provide fuel; inspect fuelTime and vehicle-detail history fallback.
Fuel loads but MPG is blank	Optional fuel-energy report may be missing, outside permissions or have invalid totals.
signal timed out / 504	Distinguish browser timeout, app deadline and Vercel FUNCTION_INVOCATION_TIMEOUT in logs.
Stops are missing	Check source issues, current source badge, viewer filters versus planner corridor and confirmed road access.
No feasible plan	Read valid-model constraints in Part I; it is different from an API failure.
Satellite has no roads	Check vector-reference readiness and raster fallback tile errors; fuel data may be healthy.
Message number ambiguous	Select the full catalog suggestion or specify brand/details manually.

17.2 Connection truthfulness

A partial fleet response keeps successful connections and warns about failed ones. Missing telemetry remains null or unavailable. Refresh failure retains last values with an unavailable/stale-status message rather than reporting a clean fresh fleet. The server warns with hashed client ID, elapsed time and endpoint/status diagnostics; its fleet log deliberately omits actual tokens and vehicle locations.

Browser Activity can include route descriptions, source names, costs and user-facing error text. The Network panel contains normalised fleet data and source information by design. Screenshots of these tools can therefore reveal operational data even when they contain no credential. The server's no-store response header stops normal HTTP caching of API responses but does not encrypt IndexedDB exports or provide user-level access control.

17.3 Credential/data separation

The Google credential, Samsara tokens and ORS key are used server-side. Public config strips Samsara tokens, and the public build excludes server setup files. Custom Sheet URLs are constrained to supported Google references. HTML-escape helpers and textContent protect normal text rendering. CSV exports apply their described formula-prefix safeguard.

These are useful boundaries, not a comprehensive security certification. The app has no per-user authentication, client entitlements, request-rate limiter shared across instances, central audit database,

content-security-policy header, or encryption of its browser cache. The sec-fetch-site guard can be absent on non-browser callers. Anyone with effective access to the shared application can ask for configured fleets and accessible sources.

17.4 Source and data quality responsibility

The app validates numeric structure, duplicates and basic coordinates but does not verify supplier price freshness, tax inclusion, diesel grade, station hours, card eligibility or whether the selected master belongs to the chosen client. `loadedAt` is an import timestamp, not a supplier update timestamp. The operator can choose another master deliberately, but the code currently cannot distinguish deliberate choice from a mistaken cross-client selection.

Manual form values, custom messages and backup records remain user-editable. Editing a driver's suggested stop after a plan does not re-run its cost calculation. A map route is advisory geometry, with standard-road and HGV limitations documented in Part I. These are concrete implementation boundaries to consider during future improvements, not newly added workflows.

Source: `server.js` `error/cache/security` handlers; `data.js` `escapeHTML` and `toCSV`; `workspace.js` `action`, `refresh`, `log` and `persistence` paths.

18. Testing, integrity and evidence

18.1 Documentation verification result

The uploaded source was copied into an isolated verification directory. `npm run build` completed successfully. `node --test tests.mjs` completed with 53 passed tests, zero failed, zero skipped, zero cancelled and zero todo. The observed duration was approximately 3.27 seconds. Node was v24.19.0 in this environment; the project target remains Node 22.x. No source changes were needed for this verification.

All 22 current file-hash references in `HYBRID-UPDATE-MANIFEST.json` matched the uploaded files. The manifest records the Hybrid update and protects previous planning/provider modules from accidental edits. The preceding `UPDATE-MANIFEST.json` is a historical page-update record. It is not an additional executable build system.

18.2 Calculation and parsing tests

The suite covers the exact full/half solver, 600 generated comparisons against exhaustive subset search, strict/rescue/end-window constraints, zero-stop and first-stop cases, independent replay, directional spurs and invalid/conflicting records. Geometry cases test projected stops between vertices, repeated passes, California entry and exclusion. Clustering/load cases check aliases, chaining and actual endpoint eligibility.

Data tests exercise quoted/multiline CSV, header aliases, invalid numbers, duplicate price conflict, preferred price columns, source GID parsing, template shapes, older message layouts and export handling. These tests validate the accepted contracts of the current parser, not all imaginable spreadsheet formats or real supplier correctness.

18.3 Page and map tests

JSDOM runs the application with a fake Leaflet implementation, fake browser workers calling the real solver, simulated APIs and in-memory storage/clipboard scaffolding. It exercises manual override protection, Use live, route search and results, message drafts/copying, reordered waypoints, hidden library access, truck visibility before/after routing, stop filter/reset rules, fleet refresh overlap, partial status and Excel worksheet selection.

Hybrid tests transform style fixtures and simulate readiness, fallback and cancellation. Release guards compare protected file hashes and selected planning/server source sections. These catch unintended model changes during presentation updates. They do not prove that real GPU/WebGL rendering, public tiles, satellite labels or every viewport look correct.

18.4 Backend integration and fault tests

The backend tests start a local Express server but mock outbound Google and Samsara fetches. They generate test credentials in memory, use synthetic tokens and verify private registry lookup, renamed tab titles/GIDs, multiple tokens, pagination, client isolation, latest-stat merge, direct MPG arithmetic and historical fuel fallback. No real provider secret is used for these tests.

Fault cases simulate one stalled connection, stalled response body, nonterminating pages, optional endpoint rejection, stalled stats and inventory authorisation failure. Application deadlines are shortened in the test fixture so these failures can be verified quickly. Additional cases cover bounded optional reports and recovery on the next refresh. A fast passing fault test therefore verifies the deadline control flow, not real waiting for 45 seconds in production.

18.5 Not verified by this run

- A real production Google key, workbook sharing configuration or live Samsara account/permissions.

- Real OSRM/ORS response correctness, station driveway snapping or commercial-vehicle road legality.
- Live map label visibility at every zoom, real satellite tile coverage or WebGL/browser GPU behavior.
- A complete Node 22 execution in this environment, full mobile/device matrix or accessibility certification.
- Provider-wide load testing, distributed cache/queue behavior, vulnerability scan or penetration test.
- Every possible route/input size or all floating-point cases beyond the targeted and generated tests.

The complete declared test-title index is in Appendix E. The source tests remain unchanged so future maintainers can run the same baseline before changing behavior.

Source: tests.mjs; HYBRID-UPDATE-MANIFEST.json; documentation verification run, 13 September 2026.

19. Current gaps and reconciliation with earlier plans

19.1 Implemented versus discussed

Earlier requirement / document statement	Current snapshot behavior
Explicit editable client-to-fuel-master mapping	Not implemented. Client and source selectors are independent.
Any telematics key can be managed through the Master	Only Samsara tokens are understood; another provider needs an adapter.
All PurFCO route/library logic participates in normal planning	Hidden library/batch code is retained, but normal Find routes uses current typed points only.
Pump/Stop naming replacement	Updated visible labels use Stops; legacy code/header/storage names remain for compatibility.
One mile by road	Geographic corridor first; directional road access is costed without a one-road-mile eligibility cap.
Arbitrary purchase quantities	Current optimiser is restricted to one full/half departure target.
All telemetry editable everywhere	Planning assumptions and compact fields are editable; raw provider metadata is not rewritten across every list/map.
Live every minute	Browser polling with caches, upstream measurement age and suspension limits; not a guaranteed continuously running scheduler.
Auto-refresh message catalogs	Registries refresh, but the 60-second loop does not reload every selected message catalog.
Default Young's forever	Fresh selection is first available; later browser preferences persist.
Original filling-instruction messages	Replaced by editable primary/team paired-stop templates; old helper remains unused by current UI.
Full/half messages match quantities automatically	Current template still asks to fill up; operators can edit, but it does not derive text from fill target.
HGV settings prove truck-safe roads	ORS fields are sent; live provider behavior and missing explicit vehicle_type need confirmation.

19.2 Documentation drift inside the repository

README.md and ENVIRONMENT.txt originated before later page and Hybrid updates. Their architecture and many setup details remain useful, but submit-only address search now coexists with Photon typeahead; PHOTON_URL is an actual variable; Routes & loads and Save Trip are hidden; the long vehicle/manual dialog and filling-instruction field are gone; the current test count is 53 rather than the earlier count described in notes.

The fuel parser accepts a valid price/latitude/longitude header set even without a name header, while some old setup text describes the first four template columns as compulsory. Latest source, current tests and current manifests take precedence when those statements differ. This manual calls out those differences instead of copying historical documentation as current behavior.

19.3 Maintenance risks visible in the source

The central workspace.js file is large and stateful, and styles.css has successive override layers. Tests protect important behavior, but cross-page changes need care because one map and source state serve several views. A raw rename of every pump identifier can break accepted CSV headers, old URLs and stored preferences even though UI wording should say Stop.

Source records can be missing or ambiguous, and configured datasets have no enforced client price entitlement. The optional reports can trigger many requests across a large fleet; the UI hydrates two at a time, but serverless instances do not share a global cache/rate budget. Historical lane labels/waypoint paths need review before restoring the hidden page. The retained Esri fallback references have an external retirement lifecycle.

These findings are an implementation inventory. No fix was applied during documentation, and none should be assumed deployed merely because it appears in a future-work discussion.

Source: cross-file review of the uploaded archive; specific modules referenced in preceding chapters.

20. Extension guide, including other telematics providers

20.1 What can remain stable

A new telematics provider should be converted into the existing internal vehicle contract before reaching the UI: stable identity, name, coordinates, GPS time, location, speed in MPH, heading degrees, fuel percent/time, optional odometer/engine metadata, MPG/source and warnings. The optimiser then continues reading manually editable gallons/MPG and requires no provider-specific formulas.

The map, Fleet list and compact Planner card can reuse that normalised contract. Google Sheets can remain the editable connection registry. Existing Samsara behavior can remain an adapter rather than being replaced. This is the extension boundary that makes future providers manageable; merely swapping a key in the current Samsara column is insufficient.

20.2 Proposed adapter design - not implemented

Layer	Future responsibility
Connection registry	Explicit client ID/name, provider, connection ID, enabled state and credential reference/configuration.
Provider adapter	Authentication, endpoint calls, paging, rate limits, timeouts, token renewal and field interpretation.
Normalisation	Units, timestamps, nulls, stable vehicle identity and provenance into Fuelio's common record.
Aggregation	Combine authorised connections, deduplicate by deliberate provider-aware identity and report partial failures.
Browser	Display/edit the common record without knowing vendor-specific response paths.
Tests	Fixture contracts, missing fields, expired credentials, pagination, two-provider collisions and recovery.

Different providers may require OAuth refresh tokens, account/database identifiers, login sessions or several credentials rather than one API key. Some may not provide fuel percent or sufficient historical totals for MPG. These must remain unavailable/editable rather than fabricated. Live polling intervals must respect each provider's actual data cadence and limits.

Provider-aware IDs are particularly important. Two vendors can both have vehicle ID 123 while describing different trucks. The current Samsara-only merge by ID is not sufficient for a future mixed-provider fleet. A stable internal vehicle ID or explicit vehicle mapping would avoid both accidental merges and duplicate map markers.

20.3 Editable client-price mapping - not implemented

A future mapping should identify which fuel masters belong to each client and an optional default, with user-editable assignments stored outside hardcoded UI lists. The application should show the actual loaded source name and handle deliberately authorised overrides explicitly. Server source requests should validate the intended mapping if preventing accidental cross-client pricing is a requirement; a dropdown default alone is not enforcement.

Adding this mapping need not alter fuel arithmetic. It changes which priced records are eligible to be supplied to the existing planner. Migration should preserve current master IDs, retained browser preferences

and the user's ability to change an assignment without a code deployment.

20.4 Change-impact map

Desired change	Primary files / tests to review
Labels, layout, theme	index.html, styles.css, workspace renderers; page/DOM tests.
Stop search/filter rules	view-model.js, workspace.js; stop map/list synchronisation tests.
Message wording/editor	messages.js, message-editor.js, message-template.csv; exact string/draft tests.
Map imagery/road labels	hybrid-map.js, workspace.js, styles.css, build.js; Hybrid lifecycle/style tests.
Fleet field display	view-model.js, workspace.js; missing-data and page tests.
Samsara endpoints/normalisation	server.js; multi-token, latest-time, timeout and report tests.
New telematics provider	New adapter modules plus server registry/aggregation; add them to server deployment as needed.
Google schema/source loading	data.js, server.js, source UI; registry/GID/import tests.
Address provider	place-search.js and server.js; stale suggestion/dismissal/coordinate tests.
Road provider/access model	server.js, geo.js, workspace planRoute/enrichCandidates; route/access and calculation tests.
Optimisation objective/rules	optimizer.js, worker contract, result export/chart, Part I proof; independent enumeration tests.
Restore hidden load tools	index.html, styles.css, workspace library/batch paths, batch.js, route-library.json; eligibility/export/UI tests.
New public file	build.js whitelist and build test, not just index.html.

20.5 Optional browser model-context tools

When `document.modelContext` is available, `workspace.js` registers four browser tools: `read_fco_workspace`, `stage_fco_trip_inputs`, `find_fco_routes` and `calculate_fco_fuel_plans`. Registration happens after bootstrap and is tied to a lifecycle signal aborted on `pagehide`. Browsers without that capability continue with ordinary UI controls.

The read tool exposes the current selected client/source, form, selected route IDs and plan summaries. The staging tool accepts only the whitelisted form fields, basic primitive/numeric/select validation and dispatches normal input events. It checks numeric lower bounds, not every later solver limit. It does not automatically find routes or send a message. The other two invoke the same route and solve functions used by the buttons. They are not public server endpoints, do not reveal API keys and do not turn Fuelio into an autonomous dispatcher. Final planner validation still applies even after staging.

Source: `workspace.js`:2362-2498. The proposed adapter/mapping sections above are design guidance, not implemented code.

21. Reference and handoff notes

21.1 Reading the two volumes together

Part I is the authority for model equations, fixed-fill exactness, route corridor/access semantics, California rules, worked examples and retained batch calculations. This volume is the authority for data acquisition, UI behavior, formats, operations and file-level maintenance. The appendices provide a complete source inventory, hashes, locked dependency paths, runtime-function locator, static control inventory and declared test index.

All source references refer to the uploaded snapshot. The full original ZIP SHA-256 appears on each cover and the current file hashes appear in Appendix A. Use that identity to distinguish these manuals from a later revision. No claim is made that an independently changed live deployment still matches this archive.

21.2 External primary references

External pages were checked only to clarify service interfaces/policies and deployment limits. They were not used as a substitute for reading the project. Their current availability, account entitlements and later changes remain outside the archive.

Reference	Used for
OSRM API	Route alternatives, coordinate order and directional Table-service semantics.
ORS routing options	HGV restrictions and the vehicle_type caveat.
Samsara snapshot	External endpoint reference for the source's snapshot integration.
Samsara fuel/energy reports	External endpoint reference for optional MPG data.
Photon project	Live-suggestion provider background.
Nominatim policy	Explicit lookup versus prohibited autocomplete and shared-service limits.
Google Sheets usage limits	Published quotas and pricing status at the documentation date.
Express on Vercel	Express function and public static-asset deployment.
Vercel function limits	Runtime/payload boundaries for the deployment discussion.
Esri legacy basemap notice	Maintenance lifecycle of reference-layer fallback services.

The source's California property cites Census TIGERweb ACS2025; the data file itself, rather than a new remote boundary extraction, was reviewed. Provider URLs are listed as documentation references, not as guarantees of unlimited access or continuing compatibility.

21.3 Handoff acceptance checklist

- Preserve the uploaded source and current hashes as the documented baseline.
- Confirm the deployed GitHub commit when comparing live behavior to this manual.
- Keep source/master identity visible when investigating price or client issues.
- Preserve null telemetry and manual override semantics when changing providers or fields.
- Preserve the special Fuel Stops highway-on/off map rule and shared Messages component.
- Treat model changes separately from interface changes and rerun independent optimiser checks.

- Recheck the live provider/deployment capabilities that the local mocked tests cannot certify.

This handoff records the existing tool in detail while making its gaps visible, so the next improvement can be specified without accidentally changing the planning or telemetry contracts.

A. Complete file inventory and integrity

All 34 files in Fuelio-main.zip were read or decoded. Text modules and documentation were examined end to end; the lockfile was parsed entry by entry; every California coordinate was traversed; the brand image was visually inspected and both SVG sources were read. No generated public/ directory, node_modules/, service-account JSON or application database is included in the source archive.

File	Bytes / lines	Responsibility
ALGORITHM.md	9,655 / 90	Earlier mathematical explanation and original planning assumptions; reconciled against the implemented solver in Part I.
batch.js	4,646 / 151	Load parsing, city connected components and directional lane eligibility.
build.js	1,841 / 61	Static public-asset whitelist, local vendor copies and third-party notices.
california.json	92,782 / 1	MultiPolygon boundary used to identify California entry and exclude physical California stops.
data.js	11,787 / 398	CSV parsing/export, numeric and header normalization, stop/message/Sheet parsing and accepted aliases.
ENVIRONMENT.txt	3,807 / 95	Environment-variable and Google service-account setup notes; does not yet list PHOTON_URL.
fuel-template.csv	70 / 1	Eight-column fuel-stop import template, header only.
fuelio-brand.png	1,483,480 / binary	2048 x 682 Fuelio brand artwork used by the header.
geo.js	5,762 / 192	Great-circle geometry, route progress, corridor candidates, coordinates and California entry.
hybrid-map.js	7,193 / 143	Transparent vector road/place overlay with raster reference fallbacks and cancellation lifecycle.
HYBRID-UPDATE-MANIFEST.json	3,606 / 112	Current presentation-update hashes plus protected source hashes.
index.html	25,118 / 651	Page shell, navigation, planner fields, views, settings and dialogs.
load-template.csv	148 / 1	Eleven-column historical-load import template, header only.
message-editor.js	16,789 / 211	Reusable editable driver/team-driver message composer and catalog stop resolver.
message-template.csv	180 / 4	Six-column message catalog template with three example stops.
messages.js	3,594 / 70	Current paired-stop message formatting and retained legacy quantitative plan-message helper.
optimizer.js	10,866 / 309	Exact fixed-fill dynamic program, validation and independent arithmetic replay.
package-lock.json	102,487 / 2728	Full npm dependency graph: 237 installed paths and one root package, with integrity metadata.
package.json	631 / 26	Package identity, Node 22 target, build/start/test scripts and direct dependencies.

File	Bytes / lines	Responsibility
place-search.js	3,485 / 56	Debounced address suggestions, keyboard selection, cancellation and stale-result protection.
README.md	21,712 / 191	Original setup, feature, architecture and troubleshooting narrative; includes some descriptions superseded by updates.
route-library.json	5,060 / 252	Twelve saved directional lane definitions; optional hidden library/batch workflows.
server.js	35,681 / 1013	Express API, private Google auth, Samsara aggregation, source loading, geocoding and road services.
styles.css	45,126 / 1800	Four themes, responsive page layouts, map pane treatment, controls, messages and later override layers.
tests.mjs	69,245 / 1655	53 passing tests covering calculations, imports, UI state, maps, messaging and mocked provider faults.
UPDATE-FUELIO.md	7,088 / 59	Page-update scope, compatibility and verification notes.
UPDATE-MANIFEST.json	2,354 / 82	Historical page-update file/integrity record.
UPDATE-MAP-HYBRID.md	4,915 / 49	Large planner map, isolated planner vehicle, dark/gold styling and Hybrid map release notes.
vehicle-rocket.svg	1,075 / 9	Direction-aware Fleet map vehicle icon, read as SVG source.
vehicle-ufo.svg	625 / 8	Direction-aware Planner map vehicle icon, read as SVG source.
vercel.json	555 / 22	Express framework build declaration and response headers; no cron configuration.
view-model.js	3,282 / 64	Pure stop filtering, vehicle visibility, heading, telemetry merge and route-highway helpers.
worker.js	243 / 8	Minimal worker request/reply wrapper around solveFuelPlan.
workspace.js	98,592 / 2498	Application state, page orchestration, map/list synchronization, live refresh, planner and storage.

The current HYBRID-UPDATE-MANIFEST.json hashes match all 22 referenced files in this archive. UPDATE-MANIFEST.json describes the preceding page update; its earlier versions are historical records rather than current-file requirements. The archive SHA on both covers identifies the complete input, including the manifest files themselves.

Source file	SHA-256 (full file bytes)
ALGORITHM.md	9ae4cba2d3bcef21880e59cf138476b1cd2df2a4b086f4ac078aceb3261cc055
batch.js	363f015035309e986ef344f4c529e65f7fa037af78328cd8ea66c9d8c2648eb0
build.js	1dd01acd8142f1a57a5e190970b77e5e99f9c9c336e9a136a21eafdc3c0157
california.json	7a9b38fa07c169af8d1dc5ab43047b414a367633ccec476c1c3ecb3c6ed9a3b5
data.js	ea8cc7a6e7080efded729554700d36b93c2aa19de98f1a70eba67de4816e2ab6
ENVIRONMENT.txt	b9521d3b582c0bcfabdc23cead8664f4322d9413525c8b58bf57fcd208b82a67
fuel-template.csv	8d871d08e038ad0c2ed1c97624f9248b0d93fd8f1a40aaf7a1fbd175805628e4

Source file	SHA-256 (full file bytes)
fuelio-brand.png	8d3537d6f3b749ea6afdc7a376c300b16bbe216e6a6f32721411ea53199585ca
geo.js	72481ad5da8e32ffd859338ec46b18c92f1a48e09b4f024f4bebc7cd629a2651
hybrid-map.js	ea8473ae10e6ce83503f4a4c37c4074ddcdf3c05579e679dceae30815948c5f9
HYBRID-UPDATE-MANIFEST.json	8ee304c60621cac886e5f0b3ca44924f8cc7c6c40202db8746f04a83f1f78949
index.html	b7a16304f33cdeb252b72b643c410e105c9dbc28beb24d15fb459e92c44e92f8
load-template.csv	55c706a803b06e87d7662dcfad25f9ee8236fdc29493608b70d354cc7b0da241
message-editor.js	bf198c50eb5ac8a56246b7cd7a72d2f84f75de08dcfe6da37e0f2cc39bdd271a
message-template.csv	e28fdf637249e2b09e1454cca5432c2b88dc2376b11b2cae80d6ae3d97645a1b
messages.js	5af21053f8520825cc16598e0c72ab349b3eceb897b821a35b0318cd807748bb
optimizer.js	bf5cb7067a8bd6d8666046029d390d50a8f01da063ef3e67b900bde3d854e610
package-lock.json	8c57b4c7d028ed9fe892a3d7c3d68a77952a064d0cfb8506935c4df352f10f61
package.json	a00df0250739ca06cdc0f528eeb4caf225fda0647ccc53f81c8a5b190f9bf61a
place-search.js	e678a4beca2d26213485b3171b81112a81eb00720dcbc9907d9443c06a35da89
README.md	b2db0bf1a3d0689ba380720f0baae8fe66955dab961384b334c982dec1e5d762
route-library.json	8e91bec88bf9447a9f8aef2296c1e6218d4840ae9bbde39f9b18bb51fcee09d8
server.js	941ae3e9e42f88a1370dde26890162f43ff32e346ba839824b2585377981914f
styles.css	a2adab6101e2e261e687583d39c42ba0166d47de7e74ebf76714b60fa3442c01
tests.mjs	ea19dbd84838b4862ee929b65d37566f8a4a46b5afefd6ff7c9aac66546a1661
UPDATE-FUELIO.md	63eaf00e0ff3421450c68c24e3e87d86edb8ca739f780c0450f20af811d514d3
UPDATE-MANIFEST.json	6607e4f3d4f5a5151b0cdb562c0200f06899a8f83f3f7b18e9f063e0af9b642e
UPDATE-MAP-HYBRID.md	511f2badf8a4a07afe880e56fc72088d6986e3a768d7aab10afb7d7e4628dfba
vehicle-rocket.svg	05f870b79b568a322e944e895d76c40e5c5988b896a36b62197c406fb18173c2
vehicle-ufo.svg	e16075d182b2e09a64110b024fee4fa6fafbb9b4d392f2898f2a2dcb58b67548
vercel.json	645712fa90aaee3a1f6bf7541347aa7645eaba6a92689710f7f9a92065cf09eb
view-model.js	467be4a5619886b077e25cd0a7d541ca72593587443df5bcc5faf18a98705634
worker.js	6a509bb6753aacbe957c71653d5ff8039fc6cfdaabfaa760227be5cacc2bd0569
workspace.js	dff78d6ace9b6c3b66d718a1681bfb15321490b2f06f38ddb92639e813027eaf

B. Complete dependency lock inventory

Lockfile version 3 contains one root package and 237 installed package-path entries: 202 without the dev flag and 35 marked development-only. Nested entries are separate installations. The following is a complete path/version/license inventory, not a vulnerability audit. License text is recorded as declared by the lockfile, not a legal interpretation.

Package path (under node_modules/)	Locked version	License / scope
@asamuzakjp/css-color	3.2.0	MIT / dev
@csstools/color-helpers	5.1.0	MIT-0 / dev
@csstools/css-calc	2.1.4	MIT / dev
@csstools/css-color-parser	3.1.0	MIT / dev
@csstools/css-parser-algorithms	3.0.5	MIT / dev
@csstools/css-tokenizer	3.0.4	MIT / dev
@fast-csv/format	4.3.5	MIT / runtime
@fast-csv/parse	4.3.6	MIT / runtime
@mapbox/geojson-rewind	0.5.2	ISC / runtime
@mapbox/jsonlint-lines-primitives	2.0.3	MIT / runtime
@mapbox/point-geometry	0.1.0	ISC / runtime
@mapbox/tiny-sdf	2.2.0	BSD-2-Clause / runtime
@mapbox/unitbezier	0.0.1	BSD-2-Clause / runtime
@mapbox/vector-tile	1.3.1	BSD-3-Clause / runtime
@mapbox/whoots-js	3.1.0	ISC / runtime
@maplibre/maplibre-gl-leaflet	0.0.22	ISC / runtime
@maplibre/maplibre-gl-style-spec	20.4.0	ISC / runtime
@maplibre/maplibre-gl-style-spec/node_modules/quickselect	2.0.0	ISC / runtime
@types/geojson	7946.0.16	MIT / runtime
@types/geojson-vt	3.2.5	MIT / runtime
@types/leaflet	1.9.22	MIT / runtime
@types/mapbox__point-geometry	0.1.4	MIT / runtime
@types/mapbox__vector-tile	1.3.4	MIT / runtime
@types/node	14.18.63	MIT / runtime
@types/pbf	3.0.5	MIT / runtime
@types/supercluster	7.1.3	MIT / runtime

Package path (under node_modules/)	Locked version	License / scope
accepts	2.0.0	MIT / runtime
agent-base	7.1.4	MIT / dev
archiver	5.3.2	MIT / runtime
archiver-utils	2.1.0	MIT / runtime
archiver-utils/node_modules/readable-stream	2.3.8	MIT / runtime
archiver-utils/node_modules/safe-buffer	5.1.2	MIT / runtime
archiver-utils/node_modules/string_decoder	1.1.1	MIT / runtime
async	3.2.6	MIT / runtime
balanced-match	1.0.2	MIT / runtime
base64-js	1.5.1	MIT / runtime
big-integer	1.6.52	Unlicense / runtime
binary	0.3.0	MIT / runtime
bl	4.1.0	MIT / runtime
bluebird	3.4.7	MIT / runtime
body-parser	2.3.0	MIT / runtime
body-parser/node_modules/content-type	2.1.0	MIT / runtime
brace-expansion	1.1.18	MIT / runtime
buffer	5.7.1	MIT / runtime
buffer-crc32	0.2.13	MIT / runtime
buffer-indexof-polyfill	1.0.2	MIT / runtime
buffers	0.1.1	Not declared / runtime
bytes	3.1.2	MIT / runtime
call-bind-apply-helpers	1.0.2	MIT / runtime
call-bound	1.0.4	MIT / runtime
chainsaw	0.1.0	MIT/X11 / runtime
compress-commons	4.1.2	MIT / runtime
concat-map	0.0.1	MIT / runtime
content-disposition	1.1.0	MIT / runtime
content-type	1.0.5	MIT / runtime
cookie	0.7.2	MIT / runtime

Package path (under node_modules/)	Locked version	License / scope
cookie-signature	1.2.2	MIT / runtime
core-util-is	1.0.3	MIT / runtime
crc-32	1.2.2	Apache-2.0 / runtime
crc32-stream	4.0.3	MIT / runtime
cssstyle	4.6.0	MIT / dev
data-urls	5.0.0	MIT / dev
dayjs	1.11.23	MIT / runtime
debug	4.4.3	MIT / runtime
decimal.js	10.6.0	MIT / dev
depd	2.0.0	MIT / runtime
dunder-proto	1.0.1	MIT / runtime
duplexer2	0.1.4	BSD-3-Clause / runtime
duplexer2/node_modules/readable-stream	2.3.8	MIT / runtime
duplexer2/node_modules/safe-buffer	5.1.2	MIT / runtime
duplexer2/node_modules/string_decoder	1.1.1	MIT / runtime
earcut	3.2.3	ISC / runtime
ee-first	1.1.1	MIT / runtime
encodeurl	2.0.0	MIT / runtime
end-of-stream	1.4.5	MIT / runtime
entities	6.0.1	BSD-2-Clause / dev
es-define-property	1.0.1	MIT / runtime
es-errors	1.3.0	MIT / runtime
es-object-atoms	1.1.2	MIT / runtime
escape-html	1.0.3	MIT / runtime
etag	1.8.1	MIT / runtime
exceljs	4.4.0	MIT / runtime
exceljs/node_modules/saxes	5.0.1	ISC / runtime
express	5.1.0	MIT / runtime
fast-csv	4.3.6	MIT / runtime
finalhandler	2.1.1	MIT / runtime

Package path (under node_modules/)	Locked version	License / scope
forwarded	0.2.0	MIT / runtime
fresh	2.0.0	MIT / runtime
fs-constants	1.0.0	MIT / runtime
fs.realpath	1.0.0	ISC / runtime
fstream	1.0.12	ISC / runtime
function-bind	1.1.2	MIT / runtime
geojson-vt	4.0.3	ISC / runtime
get-intrinsic	1.3.0	MIT / runtime
get-proto	1.0.1	MIT / runtime
get-stream	6.0.1	MIT / runtime
gl-matrix	3.4.4	MIT / runtime
glob	7.2.3	ISC / runtime
global-prefix	4.0.0	MIT / runtime
gopd	1.2.0	MIT / runtime
graceful-fs	4.2.11	ISC / runtime
has-symbols	1.1.0	MIT / runtime
hasown	2.0.4	MIT / runtime
html-encoding-sniffer	4.0.0	MIT / dev
http-errors	2.0.1	MIT / runtime
http-proxy-agent	7.0.2	MIT / dev
https-proxy-agent	7.0.6	MIT / dev
iconv-lite	0.7.3	MIT / runtime
ieee754	1.2.1	BSD-3-Clause / runtime
immediate	3.0.6	MIT / runtime
inflight	1.0.6	ISC / runtime
inherits	2.0.4	ISC / runtime
ini	4.1.3	ISC / runtime
ipaddr.js	1.9.1	MIT / runtime
is-potential-custom-element-name	1.0.1	MIT / dev
is-promise	4.0.0	MIT / runtime

Package path (under node_modules/)	Locked version	License / scope
isarray	1.0.0	MIT / runtime
isexe	3.1.5	BlueOak-1.0.0 / runtime
jsdom	26.1.0	MIT / dev
json-stringify-pretty-compact	4.0.0	MIT / runtime
jszip	3.10.2	(MIT OR GPL-3.0-or-later) / runtime
jszip/node_modules/readable-stream	2.3.8	MIT / runtime
jszip/node_modules/safe-buffer	5.1.2	MIT / runtime
jszip/node_modules/string_decoder	1.1.1	MIT / runtime
kdbush	4.1.0	ISC / runtime
kind-of	6.0.3	MIT / runtime
lazystream	1.0.1	MIT / runtime
lazystream/node_modules/readable-stream	2.3.8	MIT / runtime
lazystream/node_modules/safe-buffer	5.1.2	MIT / runtime
lazystream/node_modules/string_decoder	1.1.1	MIT / runtime
leaflet	1.9.4	BSD-2-Clause / runtime
lie	3.3.0	MIT / runtime
listenercount	1.0.1	ISC / runtime
lodash.defaults	4.2.0	MIT / runtime
lodash.difference	4.5.0	MIT / runtime
lodash.escaperegexp	4.1.2	MIT / runtime
lodash.flatten	4.4.0	MIT / runtime
lodash.groupby	4.6.0	MIT / runtime
lodash.isboolean	3.0.3	MIT / runtime
lodash.isequal	4.5.0	MIT / runtime
lodash.isfunction	3.0.9	MIT / runtime
lodash.isnil	4.0.0	MIT / runtime
lodash.isplainobject	4.0.6	MIT / runtime
lodash.isundefined	3.0.1	MIT / runtime
lodash.union	4.6.0	MIT / runtime
lodash.uniq	4.5.0	MIT / runtime

Package path (under node_modules/)	Locked version	License / scope
lru-cache	10.4.3	ISC / dev
maplibre-gl	4.7.1	BSD-3-Clause / runtime
math-intrinsics	1.1.0	MIT / runtime
media-typer	1.1.1	MIT / runtime
merge-descriptors	2.0.0	MIT / runtime
mime-db	1.54.0	MIT / runtime
mime-types	3.0.2	MIT / runtime
minimatch	3.1.5	ISC / runtime
minimist	1.2.8	MIT / runtime
mkdirp	0.5.6	MIT / runtime
ms	2.1.3	MIT / runtime
murmurhash-js	1.0.0	MIT / runtime
negotiator	1.1.0	MIT / runtime
negotiator/node_modules/content-type	2.1.0	MIT / runtime
normalize-path	3.0.0	MIT / runtime
nwsapi	2.2.27	MIT / dev
object-inspect	1.13.4	MIT / runtime
on-finished	2.4.1	MIT / runtime
once	1.4.0	ISC / runtime
pako	1.0.11	(MIT AND Zlib) / runtime
parse5	7.3.0	MIT / dev
parseurl	1.3.3	MIT / runtime
path-is-absolute	1.0.1	MIT / runtime
path-to-regexp	8.4.2	MIT / runtime
pbfb	3.3.0	BSD-3-Clause / runtime
potpack	2.1.0	ISC / runtime
process-nextick-args	2.0.1	MIT / runtime
protocol-buffers-schema	3.6.1	MIT / runtime
proxy-addr	2.0.7	MIT / runtime
punycode	2.3.1	MIT / dev

Package path (under node_modules/)	Locked version	License / scope
qs	6.16.0	BSD-3-Clause / runtime
quickselect	3.0.0	ISC / runtime
range-parser	1.3.0	MIT / runtime
raw-body	3.0.2	MIT / runtime
readable-stream	3.6.2	MIT / runtime
readdir-glob	1.1.3	Apache-2.0 / runtime
readdir-glob/node_modules/brace-expansion	2.1.4	MIT / runtime
readdir-glob/node_modules/minimatch	5.1.9	ISC / runtime
resolve-protobuf-schema	2.1.0	MIT / runtime
rimraf	2.7.1	ISC / runtime
router	2.2.0	MIT / runtime
rrweb-cssom	0.8.0	MIT / dev
rw	1.3.3	BSD-3-Clause / runtime
safe-buffer	5.2.1	MIT / runtime
safer-buffer	2.1.2	MIT / runtime
saxes	6.0.0	ISC / dev
send	1.2.1	MIT / runtime
serve-static	2.2.1	MIT / runtime
setimmediate	1.0.5	MIT / runtime
setprototypeof	1.2.0	ISC / runtime
side-channel	1.1.1	MIT / runtime
side-channel-list	1.0.1	MIT / runtime
side-channel-map	1.0.1	MIT / runtime
side-channel-weakmap	1.0.2	MIT / runtime
statuses	2.0.2	MIT / runtime
string_decoder	1.3.0	MIT / runtime
supercluster	8.0.1	ISC / runtime
symbol-tree	3.2.4	MIT / dev
tar-stream	2.2.0	MIT / runtime
tinyqueue	3.0.0	ISC / runtime

Package path (under node_modules/)	Locked version	License / scope
tldts	6.1.86	MIT / dev
tldts-core	6.1.86	MIT / dev
tmp	0.2.7	MIT / runtime
toidentifier	1.0.1	MIT / runtime
tough-cookie	5.1.2	BSD-3-Clause / dev
tr46	5.1.1	MIT / dev
traverse	0.3.9	MIT/X11 / runtime
type-is	2.1.0	MIT / runtime
type-is/node_modules/content-type	2.1.0	MIT / runtime
unpipe	1.0.0	MIT / runtime
unzipper	0.10.14	MIT / runtime
unzipper/node_modules/readable-stream	2.3.8	MIT / runtime
unzipper/node_modules/safe-buffer	5.1.2	MIT / runtime
unzipper/node_modules/string_decoder	1.1.1	MIT / runtime
util-deprecate	1.0.2	MIT / runtime
uuid	8.3.2	MIT / runtime
vary	1.1.2	MIT / runtime
vt-pbf	3.1.3	MIT / runtime
w3c-xmlserializer	5.0.0	MIT / dev
webidl-conversions	7.0.0	BSD-2-Clause / dev
whatwg-encoding	3.1.1	MIT / dev
whatwg-encoding/node_modules/iconv-lite	0.6.3	MIT / dev
whatwg-mimetype	4.0.0	MIT / dev
whatwg-url	14.2.0	MIT / dev
which	4.0.0	ISC / runtime
wrappy	1.0.2	ISC / runtime
ws	8.21.3	MIT / dev
xml-name-validator	5.0.0	Apache-2.0 / dev
xmlchars	2.2.0	MIT / runtime
zip-stream	4.1.1	MIT / runtime

Package path (under node_modules/)	Locked version	License / scope
zip-stream/node_modules/archiver-utils	3.0.4	MIT / runtime

The original lockfile also records resolved download URLs, integrity digests, dependency ranges, engine requirements, optional and peer metadata where provided. npm ci consumes those exact records. This document does not replace the lockfile, and no dependency was upgraded during documentation.

C. Runtime function and class locator

This index lists all top-level named functions/classes in the non-calculation runtime modules. Shared mathematical helpers are indexed in Part I. Arrow helpers, event callbacks and class methods remain grouped under the module contracts described in the main chapters.

File / symbol	Line	Role in source
workspace.js / dbGet	126	Internal named function
workspace.js / dbPut	135	Internal named function
workspace.js / formData	153	Internal named function
workspace.js / persist	161	Internal named function
workspace.js / toast	188	Internal named function
workspace.js / notice	194	Internal named function
workspace.js / log	201	Internal named function
workspace.js / api	206	Internal named function
workspace.js / action	224	Internal named function
workspace.js / setView	241	Internal named function
workspace.js / applyTheme	280	Internal named function
workspace.js / keyForVehicle	285	Internal named function
workspace.js / manualEdits	288	Internal named function
workspace.js / markManual	291	Internal named function
workspace.js / updateFuelUI	297	Internal named function
workspace.js / currentRules	319	Internal named function
workspace.js / routeOptions	339	Internal named function
workspace.js / tripStamp	349	Internal named function
workspace.js / updateRouteStatus	360	Internal named function
workspace.js / loadConfig	369	Internal named function
workspace.js / vehicleLabel	449	Internal named function
workspace.js / findTruck	454	Internal named function
workspace.js / loadFleet	471	Internal named function
workspace.js / updateFleet	485	Internal named function
workspace.js / applyLive	534	Internal named function
workspace.js / chooseTruck	577	Internal named function
workspace.js / displayedVehicle	600	Internal named function

File / symbol	Line	Role in source
workspace.js / fetchVehicleDetail	603	Internal named function
workspace.js / loadVehicleDetail	630	Internal named function
workspace.js / hydrateVisibleVehicles	641	Internal named function
workspace.js / renderTelemetry	662	Internal named function
workspace.js / applyFuel	685	Internal named function
workspace.js / loadMaster	708	Internal named function
workspace.js / loadCustomSheet	721	Internal named function
workspace.js / refreshFuelSource	735	Internal named function
workspace.js / uploadFuel	748	Internal named function
workspace.js / currentPumps	763	Internal named function
workspace.js / createMap	767	Internal named function
workspace.js / setMapMode	826	Internal named function
workspace.js / popup	858	Internal named function
workspace.js / visiblePumps	861	Internal named function
workspace.js / pumpsOnMap	864	Internal named function
workspace.js / drawMap	867	Internal named function
workspace.js / vehiclePopup	915	Internal named function
workspace.js / fleetRows	919	Internal named function
workspace.js / drawTruck	923	Internal named function
workspace.js / focusTruck	954	Internal named function
workspace.js / renderFleet	965	Internal named function
workspace.js / fitMap	977	Internal named function
workspace.js / focusPump	985	Internal named function
workspace.js / renderPumps	1004	Internal named function
workspace.js / geocode	1038	Internal named function
workspace.js / bindLocation	1049	Internal named function
workspace.js / searchPlace	1060	Internal named function
workspace.js / syncWaypoints	1061	Internal named function
workspace.js / restoreWaypoints	1066	Internal named function
workspace.js / renderWaypoints	1070	Internal named function
workspace.js / resolveField	1092	Internal named function

File / symbol	Line	Role in source
workspace.js / findRoutes	1099	Internal named function
workspace.js / renderRouteCards	1166	Internal named function
workspace.js / showProgress	1199	Internal named function
workspace.js / enrichCandidates	1204	Internal named function
workspace.js / solveInWorker	1282	Internal named function
workspace.js / planRoute	1311	Internal named function
workspace.js / solveSelected	1358	Internal named function
workspace.js / fuelChart	1416	Internal named function
workspace.js / renderResults	1432	Internal named function
workspace.js / download	1513	Internal named function
workspace.js / exportPlan	1522	Internal named function
workspace.js / copy	1588	Internal named function
workspace.js / initMessageEditors	1608	Internal named function
workspace.js / openPlanMessage	1624	Internal named function
workspace.js / renderMessages	1640	Internal named function
workspace.js / saveMessages	1644	Internal named function
workspace.js / messageInputs	1645	Internal named function
workspace.js / updateCustomMessage	1646	Internal named function
workspace.js / renderLibrary	1648	Internal named function
workspace.js / useLane	1671	Internal named function
workspace.js / saveTrip	1699	Internal named function
workspace.js / renderLoads	1722	Internal named function
workspace.js / runBatch	1758	Internal named function
workspace.js / renderLog	1849	Internal named function
workspace.js / renderHistory	1858	Internal named function
workspace.js / readTableFile	1883	Internal named function
workspace.js / bindEvents	1953	Internal named function
workspace.js / bootstrap	2278	Internal named function
workspace.js / registerWorkspaceTools	2362	Internal named function
server.js / cached	25	Internal named function
server.js / paced	44	Internal named function

File / symbol	Line	Role in source
server.js / fetchData	58	Internal named function
server.js / serviceAccount	112	Internal named function
server.js / googleToken	125	Internal named function
server.js / sheetMeta	162	Internal named function
server.js / sheetTable	170	Internal named function
server.js / registryRows	210	Internal named function
server.js / registry	228	Internal named function
server.js / sourceRequest	341	Internal named function
server.js / deadline	372	Internal named function
server.js / samsaraRead	399	Internal named function
server.js / samsaraPage	433	Internal named function
server.js / mergeStat	464	Internal named function
server.js / tokenFleet	483	Internal named function
server.js / publicVehicle	523	Internal named function
server.js / getClient	549	Internal named function
server.js / validPoints	765	Internal named function
server.js / orsOptions	785	Internal named function
server.js / orsRoute	805	Internal named function
data.js / number	14	Exported function
data.js / parseCSV	22	Exported function
data.js / toCSV	56	Exported function
data.js / tableRecords	153	Exported function
data.js / highwayTokens	164	Exported function
data.js / stateCode	179	Exported function
data.js / parseFuelTable	189	Exported function
data.js / parseSheetReference	317	Exported function
data.js / parseMessageTable	348	Exported function
messages.js / pumpLine	4	Exported function
messages.js / planMessage	7	Exported function
messages.js / customMessage	38	Exported function
message-editor.js / resolveMessageStop	6	Exported function

File / symbol	Line	Role in source
message-editor.js / MessageEditor	21	Exported class
view-model.js / stopNumber	4	Exported function
view-model.js / activeVehicles	7	Exported function
view-model.js / alias	11	Internal named function
view-model.js / searchStops	18	Exported function
view-model.js / stopMapRows	36	Exported function
view-model.js / headingDegrees	39	Exported function
view-model.js / mergeVehicleDetail	43	Exported function
view-model.js / routeHighways	57	Exported function
view-model.js / vehicleIconHTML	62	Exported function
place-search.js / bindPlaceSearch	2	Exported function
hybrid-map.js / makeHybridStyle	7	Exported function
hybrid-map.js / HybridReference	67	Exported class

Class methods and compact shared helpers

Member / helper	Behavior
MessageEditor.constructor	Builds shared source controls, pair controls, previews and copy/template actions; binds per-editor state.
MessageEditor.run	Disables a control during an asynchronous operation, displays its error and restores the control.
setMasters / setFallback / catalog	Refresh master choices; supply loaded-fuel fallback; choose explicit message rows or fallback rows.
applySource / updateSource	Validate a nonempty catalog, reset pairs for the new source and render its name/count badge.
renderSuggestions	Build a datalist of distinct formatted stop lines from the active catalog.
renderPairs / fillDetails	Render editable A/B pairs; bind number resolution, detail overrides and removal; synchronise small detail fields.
inputs	Ignore unused blank pairs; reject ambiguous, unresolved or incomplete manual records; return driver/partner/route/pairs.
update	Regenerate both templates, report validation failures, enable/disable copy controls and request draft save.
save / snapshot	Notify the workspace using version-2 rows/source/master/pairs/fields/previews.
restore	Accept current and older field/pair forms; validate before restoring editable previews.
setPlan	Use plan primary stops, blank alternatives, editable route/names and only compatible catalog enrichment.
HybridReference.constructor	Create noninteractive fallback/vector panes at z-index 300 and 350.
HybridReference.remove / hide	Remove active layers, advance generation, abort pending style request and clear readiness timers.
HybridReference.showFallback	Create/add two raster reference layers and report the first visible tile error.
HybridReference.show	Load/reuse transformed style, add MapLibre, watch readiness/errors and fall back if necessary.
clean / header / escapeHTML	Trim/stringify; normalise column names; escape HTML-sensitive characters.
findCol / state aliases	Choose the earliest accepted header alias and map exact state names/abbreviations.
normal / alias	Prepare search text and supplementary Mt/St/direction/Mc variants without state-query rewriting.
emptySide / emptyPair / clone	Create isolated editable message pair state and JSON-copy a saved draft.
money / fmt / time / uiText	USD formatting; finite numeric display; hours/minutes; visible legacy Pump-to-Stop wording conversion.
keyForVehicle / manualEdits	Locate a vehicle/manual-trip override bucket without altering upstream records.
hash / active / latest	Create truncated SHA identities; interpret registry activation; choose newest stat observations.
coordinateString / road	Serialise longitude-latitude pairs and queue road-provider work.
Worker onmessage	Receive one input, invoke solveFuelPlan and return success/error to the controlling browser.

D. Full HTML control inventory

Element ID / type	Default / options	Label or purpose
theme / select	light: Light; dark: Dark; crimson: Crimson; icy: Icy	Colour theme
settingsBtn / button		⚙️
client / select	(blank): Manual planning	Client
master / select	(blank): Choose a source	Fuel master
loadFuel / button		Load prices
sourceBtn / button		Sheet / file
resetTrip / button		Clear
truck / input text	Search truck number	Truck or enter a unit manually
useLive / button		Use live
origin / input text	City, state or latitude, longitude	Origin
destination / input text	Where is the truck headed?	Destination
waypoints / textarea		[hidden]
addWaypoint / button		+ Add waypoint
mapClick / select	none: Map selection off; origin: Origin; destination: Destination; waypoint: Add waypoint	Set by clicking the map
capacity / input number	240	Tank capacity gal Range: 1 to -.
mpg / input number	7	Fuel economy MPG Range: 0.1 to -.
startFuel / input number	100	Starting fuel gal Range: 0 to -.
fuelPercent / input number	41.67	Starting fuel % Range: 0 to 100.
fill / select	full: Full Tank - leave at 100%; half: Half Tank - leave at 50%	At each fuel stop
reserve / input number	50	Buffer fuel gal Range: 0 to -.
endFuel / input number	50	Ending fuel target gal Range: 0 to -.
mode / select	buffer: Plain Buffer - lowest fuel cost; strict: Strict arrival window; before_ca: Before entering California	Refuelling rule
maxStops / input number	3	Maximum stops Range: 0 to 1000.
corridor / input number	1	Route radius mi Range: 0.05 to 10.
windowLower / input number	10	Window below buffer gal Range: 0 to -.

Element ID / type	Default / options	Label or purpose
windowUpper / input number	5	Window above buffer gal Range: 0 to -.
endTolerance / input number	No upper limit	End-target tolerance gal Range: 0 to -.
rescue / input checkbox		Offer upper-window rescue if needed
findRoutes / button		Find routes →
saveRoute / button		Save trip [hidden]
streetBtn / button		Map
satelliteBtn / button		Satellite
heatBtn / button		Price heat
fitBtn / button		Fit view
fullscreenBtn / button		⌵
solveRoutes / button		Fuelio Hunt ↗ [disabled]
cancelSolve / button		Cancel
refreshFleet / button		Refresh now
fleetSearch / input text	Search truck, VIN or location	
fuelExport / button		Export CSV
pumpSearch / input text	Search stop, city, state, highway or exit	
stateFilter / select	(blank): All states	All states
priceSort / select	master: Master order; low: Lowest price; high: Highest price	Master order Lowest price Highest price [hidden]
routeOnly / input checkbox		Selected routes only
clearStops / button		Clear
highwaySearch / input text	Find a highway	
clearHighwaySearch / button		×
clearHighways / button		Clear highways
cyclePrice / button		Price / gal ↕
exportLibrary / button		Export saved routes
loadUpload / input file		Import loads [hidden]
clusterRadius / input number	50	City cluster radius Range: 0.1 to -.
prepareLoads / button		Prepare & match
runBatch / button		Plan eligible loads
exportLoads / button		Export results

Element ID / type	Default / options	Label or purpose
clearLog / button		Clear log
sheetUrl / input text	https://docs.google.com/spreadsheets/d/...	Google Sheet URL
sheetGid / input text	0	Tab GID optional if present in the URL
loadCustomSheet / button		Load Google Sheet
fuelUpload / input file		Upload fuel file [hidden]
routingProfile / select	driving: Standard road routing; hgv: Truck routing - requires configured ORS key	Road profile
truckHeight / input number	4.1	Height metres Range: 0 to -.
truckWeight / input number	36	Gross weight tonnes Range: 0 to -.
truckWidth / input number	2.6	Width metres Range: 0 to -.
truckLength / input number	22	Length metres Range: 0 to -.
hazmat / input checkbox		Hazardous goods
avoidTolls / input checkbox		Avoid tolls (truck routing)
exportBackup / button		Export workspace backup
restoreBackup / input file		Restore backup [hidden]
refreshConfig / button		Reload master registry
workbookSheet / select		Worksheet
workbookUse / button		Use this worksheet

Dynamic controls created by MessageEditor and waypoint rows are not static index.html nodes. Their complete behaviors are described in the Messages and Planner chapters. A hidden attribute on a parent can also hide controls whose own row above has no hidden flag.

E. Complete test case index

Line	Test case title
41	Hybrid references preserve imagery transparency and readable roads, places and US route numbers
69	Hybrid mode switches discard late loads and keep every reference below fuel-stop markers
92	Hybrid retains road/place references when vector rendering fails or is unavailable
109	Release guard: optimiser, routing maths, imports, libraries and Samsara code are unchanged
201	Half Tank: arrive with 119 gallons, purchase 1 and depart with exactly 120
214	Starting above half capacity does not drain the tank or force an unnecessary stop
225	Infeasible Half Tank has a Full Tank warning and no exportable stops
236	Zero stops is a real constraint; below-buffer starts do not silently relax it
254	Unreachable cheapest pump cannot enter a solution
267	Both inbound and return road access consume fuel and count in purchased cost
279	Fixed filling policy and ending tolerance are enforced together
287	Strict-window rescue retains the lower reserve, stop limit and fill target
309	Same-mile pumps remain distinct price alternatives
318	Invalid, negative, NaN and conflicting records fail loudly
338	Independent replay detects corrupted accounting
389	600 random instances match exhaustive global minimum over every stop subset
432	Fuel Stops search aliases expand normal matching without state-name expansion
451	Highway OFF uses all map records; Highway ON uses Search AND State AND highway OR
466	Fuel display merges only missing snapshot values and keeps live zeros and original records
478	Driver and partner messages match Div's wording, stars, OR lines and blank highway fields
498	Autocomplete rejects stale replies, hides on dismissal and returns on typing
527	Message editor migrates old drafts, permits manual stops, and isolates plan edits
550	CSV handles BOM, quoted commas, newlines, aliases and non-positive price rejection
564	CSV exports protect formula-leading text and retain numeric negatives
570	Sheet GID 0 is valid; wrong domains and nonnumeric GIDs fail
582	Fuel duplicate conflicts never silently select another price
590	Message brands and pump IDs stay distinct; Half Tank text agrees with the plan
612	Message master import accepts prefaced headers and original headerless layouts
629	Supplied eight-column fuel format retains coordinates, prices, highways and exits

Line	Test case title
646	Supplied six-column message format uses Location, Highway and Exit without coordinate shifts
662	One-mile corridor uses distance to any segment, including between vertices
691	Route revisits can expose the same pump at separate route positions
711	California boundary, outside start, first entry and final leg are checked
742	City clusters are descriptive; chained clusters cannot grant lane eligibility
758	Historical load aliases and manual per-load overrides are read correctly
768	Build output contains the app and excludes server code and credential instructions
794	DOM workflow: editable live values, manual overrides, themes, routes and verified plan messages
1111	Planner and standalone message editors share fields, formatting, templates and editable alternatives
1128	Fuel-only edits reuse routes; truck and waypoint edits require Find routes
1145	Deactivated trucks are hidden; fleet history fuel, MPG and map selection use the same records
1168	Fuel Stops list-only search, marker selection and price sort/reset preserve the master and theme
1192	Manual and scheduled fleet refreshes share one pending request
1208	Registry refresh does not claim live data while a partial fleet refresh waits
1260	Backend: private registry, multiple tokens, pagination, client isolation and history fallback
1524	Autocomplete uses a separately cached provider and rejects invalid/non-US responses
1601	Fleet fault: \${mode} returns a bounded, truthful result
1625	Optional vehicle reports stop at their total deadline and keep manual fallback available
1635	Fleet recovers on the next refresh after upstream failure

Parameterized fleet fault tests expand one source declaration into six runs. The Node test runner reports 53 passed tests overall. The exhaustive optimiser comparison executes 600 generated inputs within one declared test. Both counts describe different levels of the suite.